

If you position the cursor in a message row and then press the F1 key, EPLAN displays the online help for the corresponding message number, lists the possible cause(s), and suggests one or more solutions for the selected message.

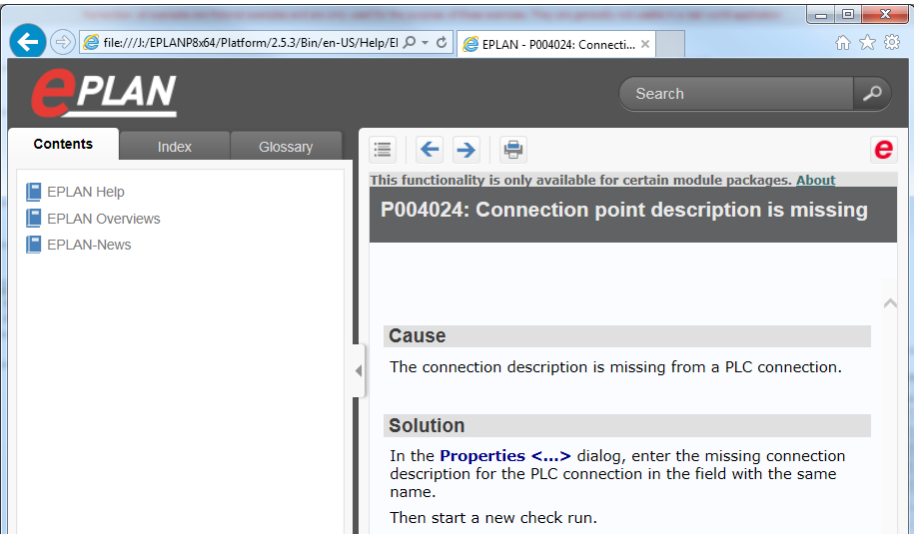


Fig. 7.30 Note on message in online help

7.2.2 Project checks

EPLAN generally distinguishes between **offline checks**, **online/offline checks** and the **Prevent errors** check type, and also allows you to “remove” checks entirely from the check run, i.e. to switch off the check with the *No* check type. You should not use this option very much since you run the risk of certain true project editing errors not being discovered (e.g. duplicate DTs) until the actual plant is built, which will present a much bigger problem.

Type of check
Online / offline
Online / offline
No
Offline
Online / offline
Prevent errors

Fig. 7.31
Types of check

The user decides what type of check routines to use. The **online check routine** is surely the first choice, when available, but is also the most computing-intensive because all the data and resulting messages must be continually checked and updated. This, of course, may have a negative effect on the performance during project editing. The **offline check routine** is the “second choice”, but I prefer to use this one. After all, who manages to constantly correct all the online messages (errors, warnings or notes) while editing a project and also working on the actual planning and design?

Check runs are started via the **PROJECT DATA/MESSAGES/CHECK PROJECT** menu or directly via the *Check project* function in the opened message management via the right-click popup menu.

7.2.3 Message classes and message categories

EPLAN allows the user to define the categories that messages belong to. This has the great advantage of allowing company-specific check runs to be developed that exactly match the way in which projects are developed within the company.

Row	Number	Message text	Category	Type of check
1	001001	Terminal without designation.	Warning	Online / offline
2	001005	Terminal of the same name use...	Warning	Online / offline
3	001008	A terminal connection point has...	Warning	Online / offline
4	001010	Terminal strip definition missing	Note	Online / offline
5	001011	Terminal strip without terminals.	Warning	Online / offline
6	001012	Incorrect saddle jumper formati...	Error	Online / offline
7	001015	Too many wire jumper levels.	Warning	Online / offline

Fig. 7.32 Example of check run message

Not all messages generated or supplied by EPLAN are, for example, true error messages. The message <<017003 Incorrect identifier>> is one example. It can be an error message, a warning, or a note, depending on the project editing sequence. You can create your own schemes for check runs in message management so that these messages are adjusted to exactly suit your project editing process.

Use the **PROJECT DATA/MESSAGES/CHECK PROJECT** menu to open the **CHECK PROJECT** dialog. In this dialog, you can select a scheme in the *Settings* field for the subsequent check run.

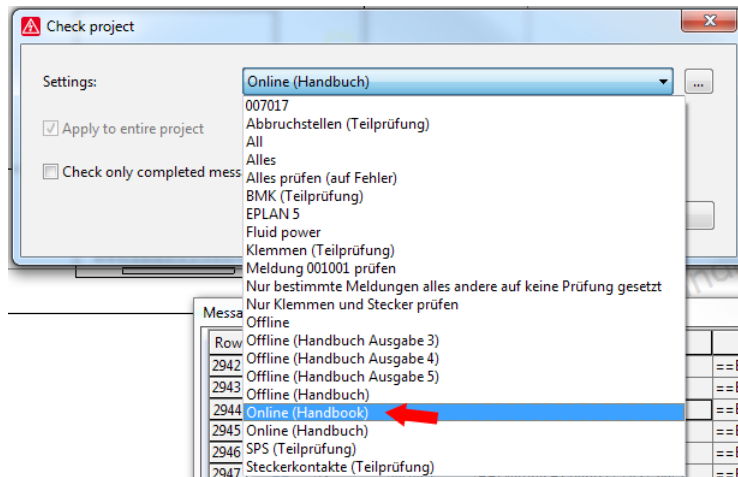


Fig. 7.33 Check project dialog

To add your own scheme to this selection, you click the  selection button. The **SETTINGS/MESSAGES AND CHECKS** dialog is displayed. To create a new scheme, you click the **[NEW]** button in the upper toolbar.

Settings for messages and check runs

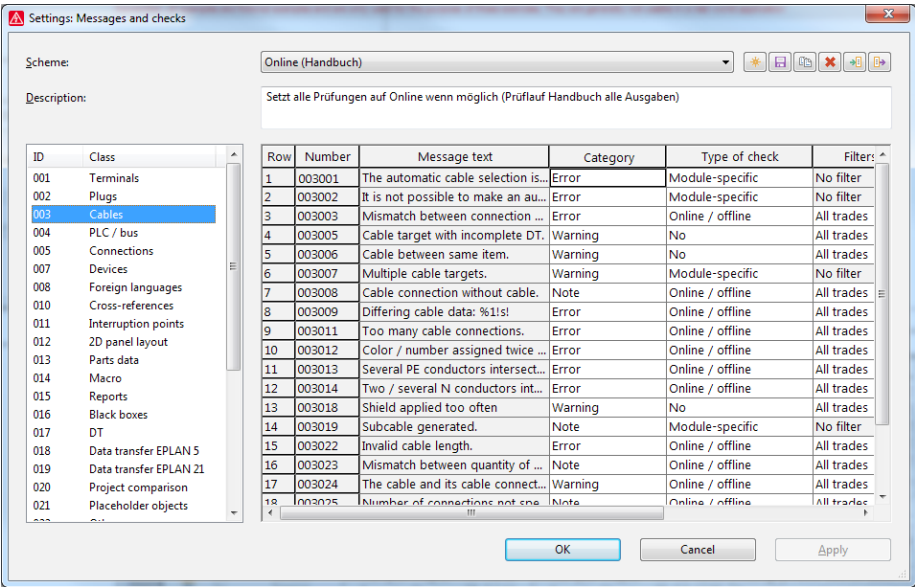


Fig. 7.34 Dialog for creating test schemes

EPLAN opens the **NEW SCHEME** dialog. Here you assign a suitably descriptive **Name** and **Description**. **Note:** The description can also be changed later. When you click the **OK** button, the scheme is saved and automatically entered into the *Scheme* selection field of the **SETTINGS/MESSAGES AND CHECKS** dialog.

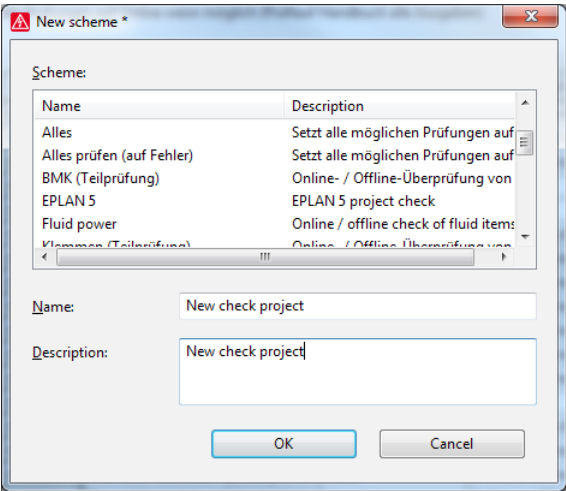


Fig. 7.35 Creating a new scheme

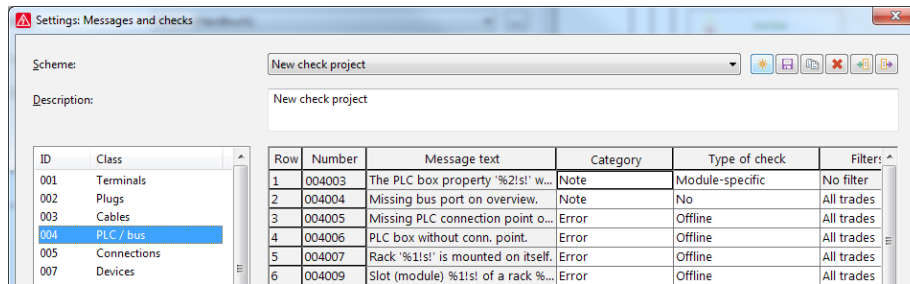


Fig. 7.36 Example of a new scheme

The individual messages can now be set and edited according to the desired specifications.

Every message belongs to a message class. There are currently 31 message classes. They range from message class 001 Terminals to message class 999 External. The message classes cannot be extended or changed; they are permanently defined by EPLAN.

Class number (ID)	Message class (class)
001	Terminals
002	Plugs
003	Cables
004	PLC/bus
005	Connections
007	Devices
008	Foreign languages
010	Cross-references
011	Interruption points
012	2D panel layout
013	Parts data import
014	Macro export
015	Reports
016	Black boxes
017	DT
018	Data transfer EPLAN 5
019	Data transfer EPLAN 21
020	Project comparison
021	Placeholder objects
022	Other
023	PPE
024	Fluid power
025	Project settings

Class number (ID)	Message class (class)
026	3D mounting layout
027	Topology
028	Pre-planning
029	Pre-planning depth
030	Subprojects
032	Drilling pattern
501	Part master data
999	External

This means that every message begins with a specific three digit number. This defines the message class. In our example, the message <017003 Incorrect identifier> is assigned the DT message class (017) and check run message 003. The rest of the message number is a three-digit number.

Every message number has a corresponding message text. As with the message number, this message text is defined by EPLAN and cannot be changed. However, in order to return to their own scheme, users can set the category and type of check for this message.

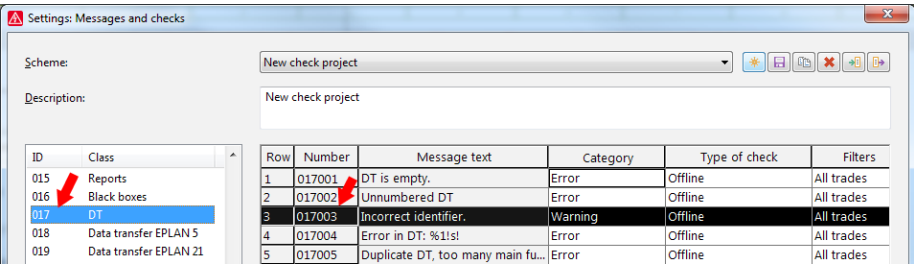
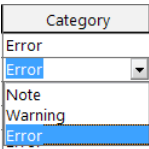
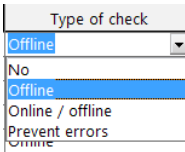


Fig. 7.37 Message text

EPLAN divides messages into three categories. There are **Notes**, **Warnings**, and **Errors**. The user must now decide on one of these three message categories. Once the category for the message has been defined, a check type must be decided. There are several check types in EPLAN.



The **No** type of check does not carry out any checks for this message either automatically or manually. This will be clarified with a short example.



Row	Number	Message text	Category	Type of check	Filter:
1	001001	Terminal without designation.	Warning	No	All trades

Fig. 7.40 Type of check set to “No”

Fig. 7.38
Category

Fig. 7.39
Types of check

A terminal will be positioned in the schematic. This terminal will not given a designation, and the check for the message *Terminal without designation* will be set to **No**. If a check run is now carried out, there will not be any check run message *Terminal without designation*. **Note:** The *Select* check box filter was set to active. Now, only messages are displayed that concern the highlighted selection.

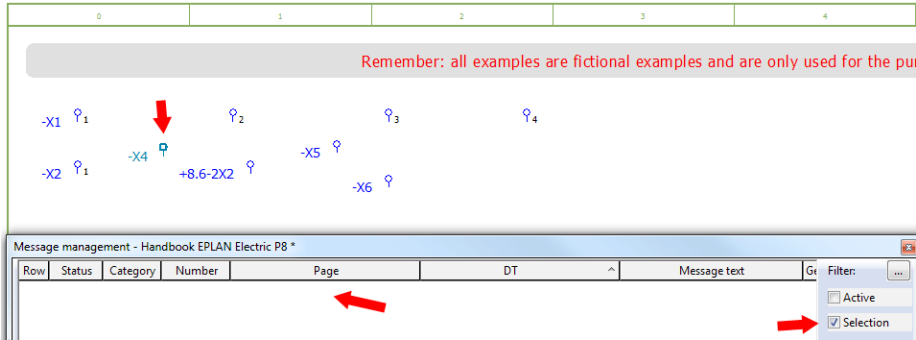


Fig. 7.41 Message management after completed check run

The check type **Offline** only generates messages when a check run is manually started.

Row	Number	Message text	Category	Type of check	Filter:
1	001001	Terminal without designation.	Warning	Offline	All trades



Fig. 7.42 Type of check set to “Offline”

Now, a terminal will again be positioned in the schematic. This terminal will also not be given a designation, and the check for the message *Terminal without designation* will be set to **Offline**. EPLAN does not yet check this terminal during placement. Only when the check run has been started manually (offline) does EPLAN generate the check run message *Terminal without designation*.

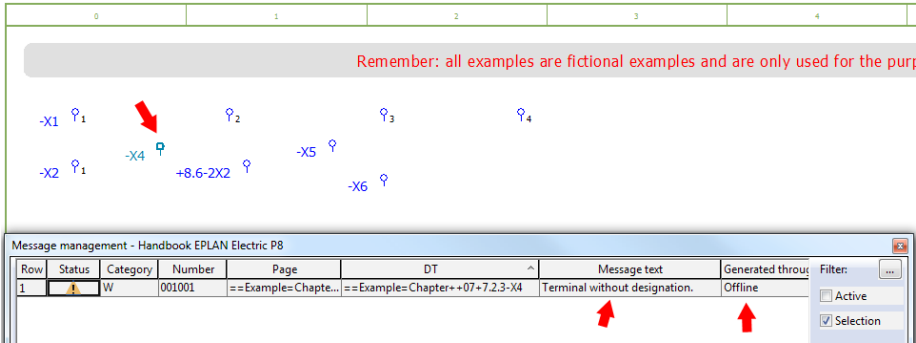


Fig. 7.43 Type of check set to “Offline”

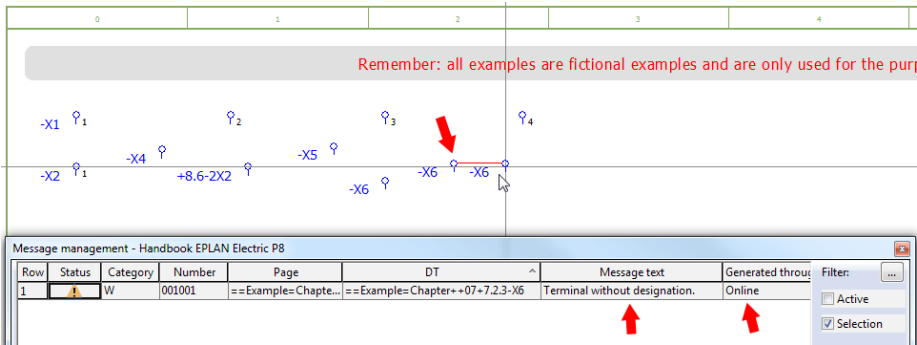
The check type **Online/offline** generates messages online and immediately enters them into the opened message management. Completed messages are not removed from the message database until the next check run.



Row	Number	Message text	Category	Type of check	Filter:
1	001001	Terminal without designation.	Warning	Online / offline	All trades

Fig. 7.44 Type of check set to “Online/Offline”

Another terminal is positioned in the schematic. This terminal is not given a designation, and the check for the message *Terminal without designation* has been set to **Online/offline**. In contrast to the previous check runs, EPLAN checks whether a designation has been entered already during the placement of the terminal. Since in our example no designation was assigned, EPLAN generates immediately (online) the corresponding check run message *Terminal without designation*.



The screenshot shows a schematic diagram with several terminals labeled -X1, -X2, -X4, -X5, -X6, and -X6. A red arrow points to the second -X6 terminal. Below the schematic is the 'Message management - Handbook EPLAN Electric P8' window. It displays a table with the following data:

Row	Status	Category	Number	Page	DT	Message text	Generated through	Filter:
1		W	001001	==Example= Chapte...	==Example= Chapter+ +07+7.2.3-X6	Terminal without designation.	Online	All trades

Red arrows point to the 'Terminal without designation.' and 'Online' columns in the message management table.

Fig. 7.45 Type of check set to “Online/offline”, EPLAN reports the error immediately

The **Prevent errors** type of check means that EPLAN checks the input during an action, for example, when inserting devices, and, if this results in errors, shows a message immediately and then “prevents” the action.



Row	Number	Message text	Category	Type of check	Filter:
1	001001	Terminal without designation.	Warning	Prevent errors	All trades

Fig. 7.46 Type of check set to “Prevent errors”

Again, a terminal is inserted and placed without a designation. During the placement, EPLAN checks whether this action violates any check and immediately reports the error in the **PREVENT ERRORS** dialog.

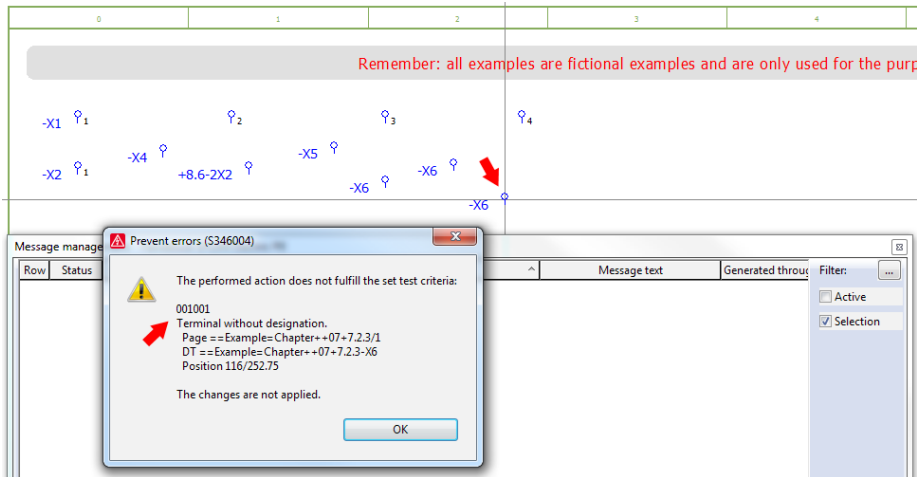


Fig. 7.47 EPLAN prevents the error on the basis of the type of check defined

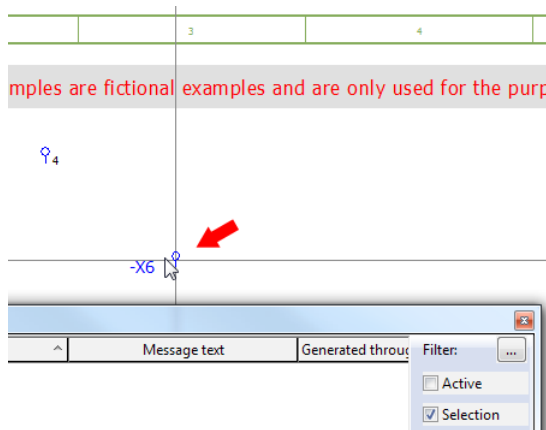


Fig. 7.48 Terminal can be repositioned

NOTE: The **Prevent errors** type of check, of course, is very strict and should (in my personal estimation) not be used during ongoing project editing. But when, for example, schematics are revised, this is a very good opportunity to immediately prevent errors while revising schematics instead of correcting them later during a check run.

EPLAN cancels this action, and you can reposition and label the terminal. If the specifications are met (terminal designation exists), the terminal can be placed.

The **Module-specific check** type of check also generates messages during certain automated actions, but cannot be influenced any further.

In addition to message management, other modules also carry out checks in EPLAN and then generate messages in the MESSAGE MANAGEMENT dialog. These checks are displayed in the MESSAGE MANAGEMENT dialog, in the *Type of check* column, as “module-specific”. If a module-specific check is not to be run, the type of check for this message must be set to “No”.

Row	Number	Message text	Category	Type of check	Filters
1	003001	The automatic cable selection is...	Error	Module-specific	No filter

Fig. 7.49 Module-specific check

If you do not want that, set this type of check to **No**.

7.2.3.1 Filters for all check run messages

Besides the previous settings for types of check or categories, it is possible to define a separate filter for each check run message.

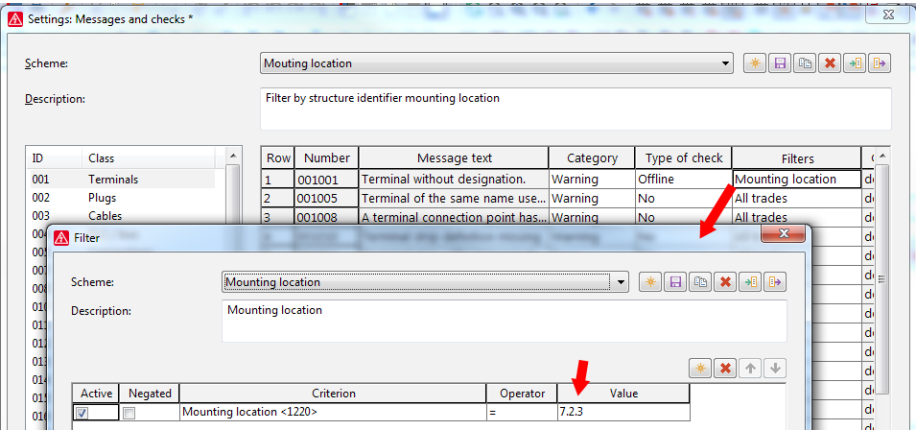


Fig. 7.50 Filter options for each check run message

These filters allow you to limit certain checks to specific areas or properties. In other words, checks can be targeted at specific objects..



The *Terminal without designation* check is set to the **Offline** type of check, but is to be checked only in a specific mounting location. This will exclude from the check all other faulty terminals that do not match the defined filter criterion.

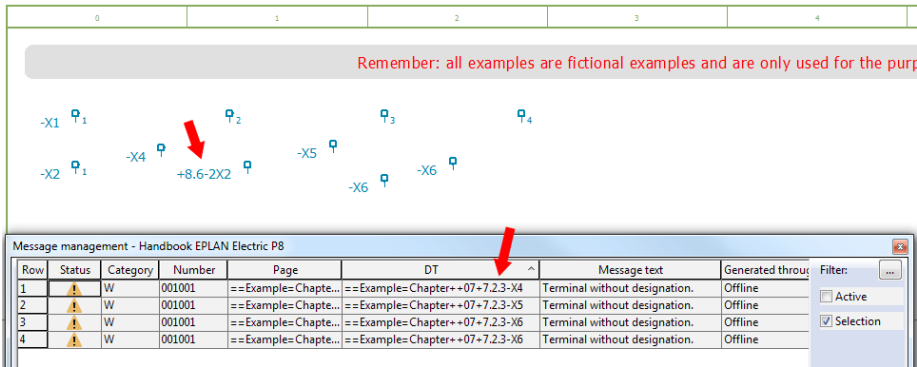


Fig. 7.51 Only checks defined by filter will be carried out

In the example, the terminals without a designation, whose mounting location was different than the one defined in the filter, were ignored by the check run. The terminals whose mounting location matches the filter, however, were captured by the check run, and EPLAN generated a corresponding check run message.



TIP: This feature (placing filters on individual check run messages) should be used very sparingly. The check run could be slowed down considerably as a result, because EPLAN would always have to evaluate the check run message as well as the filters.

7.2.4 Filters in message management

As described in the section 7.2.3, EPLAN messages are categorized into specific message classes. All message classes and areas (electrical engineering, fluid, or e.g. errors) can be switched on and off via filters.

7.2.4.1 Filter – Active setting

Select the **ACTIVE** check box and click the  button to open the **FILTER: MESSAGES** dialog.

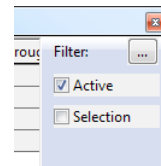


Fig. 7.52
Filter – Active

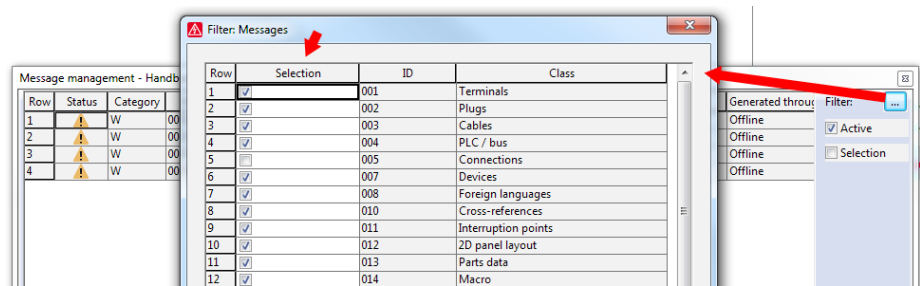


Fig. 7.53 Filter settings for message class selection

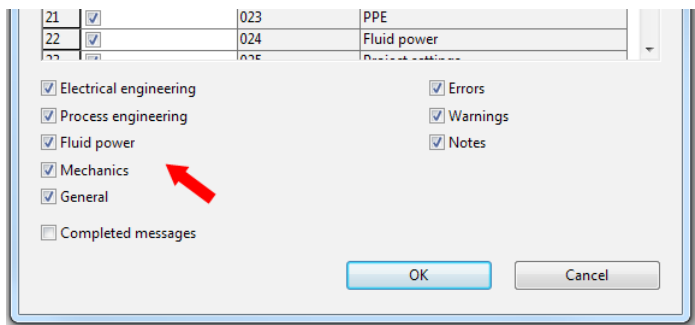


Fig. 7.54 Filter settings for the areas

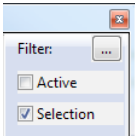
In this dialog, by deselecting the **message classes** (terminals, plugs, cables, etc.) and also the **areas** (electrical engineering, process engineering, etc.), it is possible to filter different data. When you are finished making entries, you close the dialog by clicking the **OK** button. To activate the filter for the message management display, you must set the check mark in the **Active** check box below the filter (see Fig. 7.55 in section 7.2.4.2).

A typical example of a filtered display would be, for example, removal of the completed messages from message management until the next check run.

7.2.4.2 Filter – Select setting

If the **SELECT** check box is set, then only messages from selected objects are displayed in message management. All other messages from other objects, etc. are hidden. This avoids tedious searching for messages relating to particular objects.

Fig. 7.55
Filter – Select



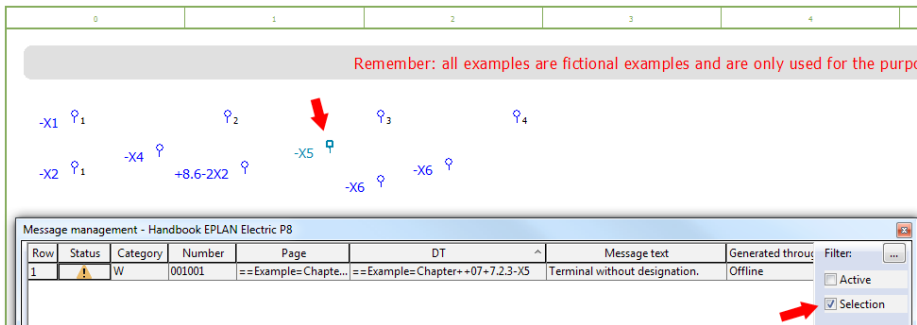


Fig. 7.56 Selection has been activated

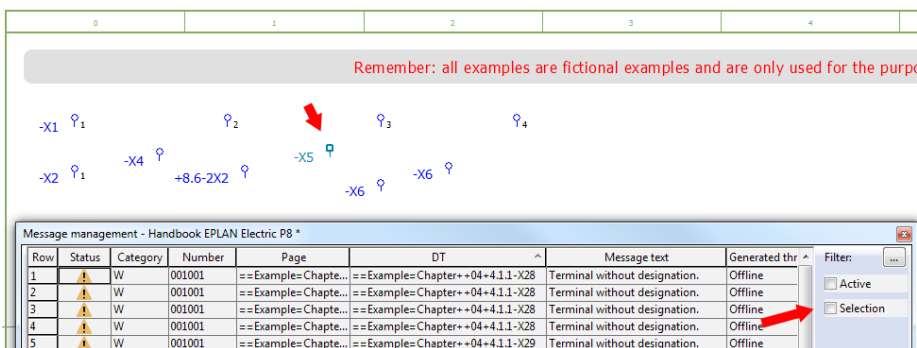


Fig. 7.57 Selection has not been activated

If the **SELECTION** check box is not active, message management displays all messages. A selection of objects does not play any role here.

7.2.5 Various message edition options

This section does not deal with the correction of generated error, warning, or note messages. I refer you instead to the online help, which explains problems and related solutions for check run messages on a much wider scale.

Instead, this section focuses more on the ways in which messages can be localized in the schematic, and what tools EPLAN provides for this. The general functions for displaying, for example, Freeze panes or Adjust column width as well as copy functions will not be explained any further here.

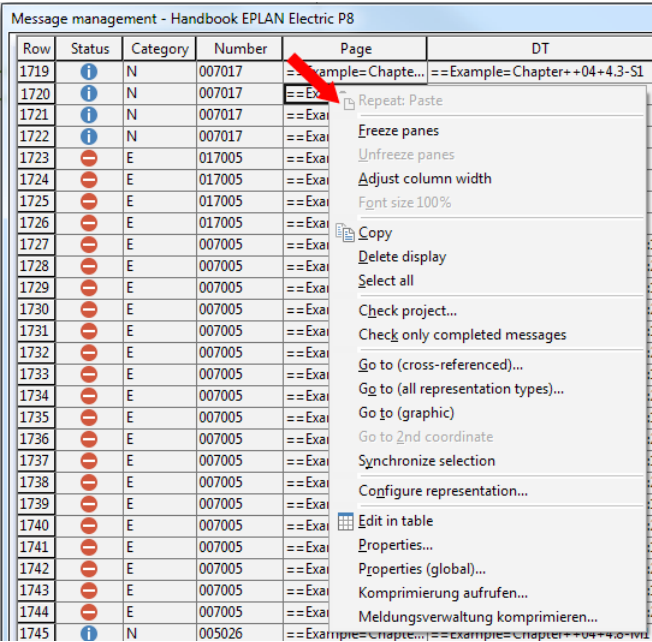


Fig. 7.58 Popup menu in the Message management dialog

The right-click popup menu provides several options/functions for easily finding errors. The key ones are described below.

- **DELETE DISPLAY:** This option deletes all messages from the message management display. The messages are still present, but are no longer displayed. They re-appear after the next check run.
- **CHECK PROJECT:** This function carries out the set check run.
- **GO TO (CROSS-REFERENCED):** This function causes EPLAN to enter all cross-referenced objects in the *Go to* list and then open the **FIND** dialog with the *Go to* tab.
- **GO TO (ALL REPRESENTATION TYPES):** This function determines all representation types (schematic, report pages, etc.) and also enters them in the *Go to* list. From here, you can navigate directly to the desired representation.
- **GO TO (GRAPHIC):** This function allows you to navigate to the object in question directly from message management. **Note:** Double-clicking a selected check with the left mouse button yields the same result. EPLAN jumps directly to the (checked) object in question (and opens the page).
- **GO TO 2ND COORDINATE:** If, for example, there are two duplicate terminals, you can use this function to jump directly to the other terminal.
- **SYNCHRONIZE SELECTION:** This function, familiar from other navigators, lets you jump to and select exactly this device in other open navigators, such as the device navigator. This function eliminates time-consuming searching for a device in the navigators.
- **CONFIGURE REPRESENTATION:** This function allows the number of columns displayed in message management to be increased or decreased.

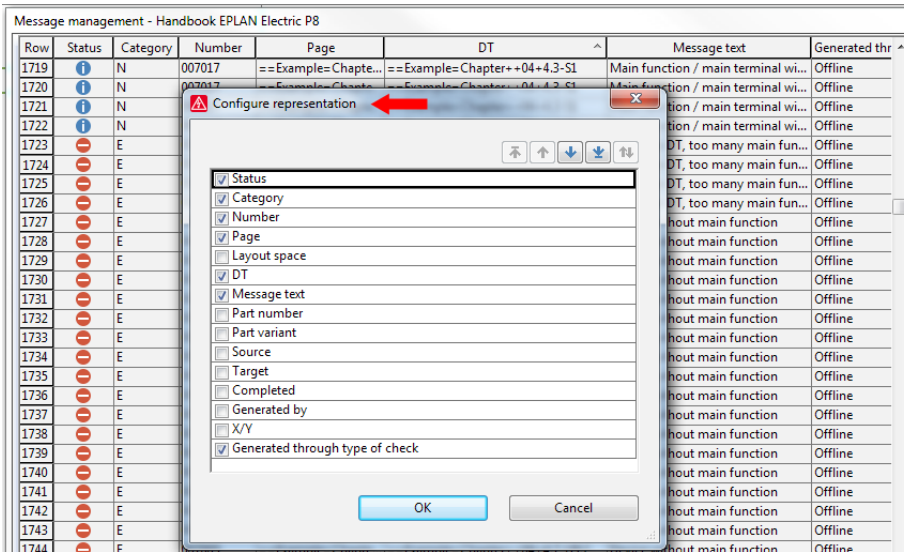


Fig. 7.59 Configure representation

When this function is executed, EPLAN displays the **CONFIGURE REPRESENTATION** dialog. Here you can remove or add columns, or change the sequence of the columns.



TIP: To obtain a preview of the incorrect objects, the **GRAPHICAL PREVIEW** should be enabled in the **VIEW** menu. Ideally, this preview should be used with a multi-screen system.

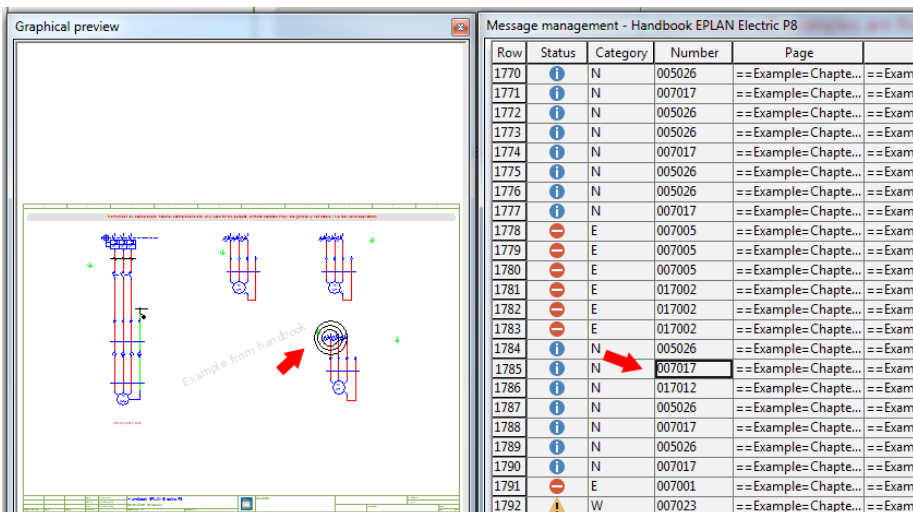


Fig. 7.60 Using the graphical preview to find errors

7.3 Layer management

What are layers? Layers are an essential element originating from mechanical engineering (CAD). Information of the same type (e.g. dimensions) is placed on the same layer. For example, the form, color, font size or other formats of this layer can later be changed at a central place, easily and without errors.

This is a great advantage, because otherwise every property that is not allocated to a layer must be manually edited. Layer management makes such editing actions a “piece of cake”. EPLAN has taken this well-known idea from mechanical engineering and extended it into the CAE area, and developed it further from one version to the next.

You start layer management via the **OPTIONS/LAYER MANAGEMENT** menu. The **LAYER MANAGEMENT** dialog is displayed. Layer management itself is a project-specific setting, and you can recognize this from the window title bar that displays the currently active project name. The dialog has windows on the left and right sides. The left window contains a tree with the superior layer designations (nodes and any subnodes) and then the layer designation **EPLANxyz**, and the right window contains the associated sub-entries (the actual layer information).

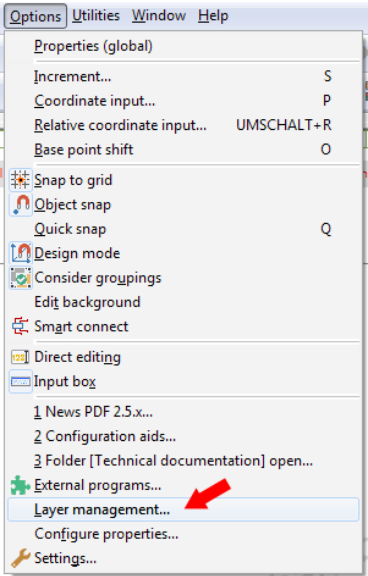


Fig. 7.61
Layer management

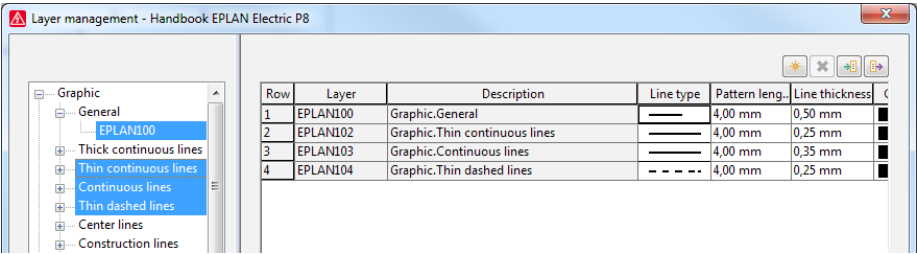


Fig. 7.62 Layer management dialog

7.3.1 Standard layers

EPLAN comes with a number of standard layers. They begin with the name EPLAN, followed by a three-digit number.

 **NOTE:** Original EPLAN layers cannot be removed (deleted).

Row	Layer	Description	Line type	Pattern leng.	Line thickness	Color	Font size	AI
8	EPLAN306	Symbol graphic.PLC boxes	- - - - -	4,00 mm	0,25 mm		2,50 mm	Low
9	EPLAN307	Symbol graphic.Structure boxes	- - - - -	4,00 mm	0,25 mm		2,50 mm	Low

Fig. 7.63 Standard EPLAN layers

7.3.2 Export and import of layers

All layers can be modified and/or adapted to your requirements, including the original EPLAN layers. However, you should first export all layers before making any changes to the standard layers. You then have a functioning backup copy of the previous layer settings.

Layers are exported as follows: Before starting the export, you must first select all desired layers using the usual Windows functions. The upper right of the dialog contains the graphical buttons. Use the  button to start the export. EPLAN opens the **LAYER EXPORT** dialog. The file name contains a default value set by EPLAN, which is the name of the project. When you confirm the dialog by clicking **SAVE**, the layers are exported and saved.

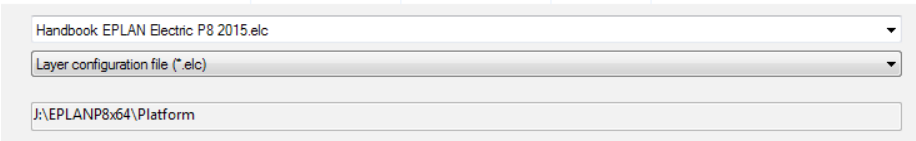


Fig. 7.64 Layer export

Layer import is done in a similar way. Clicking the  button causes EPLAN to open the **LAYER IMPORT** dialog. Here you select the desired layer configuration file and import it into the opened EPLAN project by clicking the **OPEN** button.

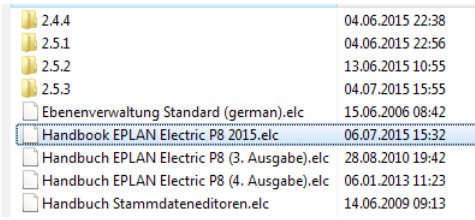


Fig. 7.65 Layer import



NOTE: All layers not already present in the project are imported and created as new layers. If layers of the same name already exist in the project, then the data in the import file always has priority. This means that layers in the project might be updated.

7.3.3 Create and delete your own layers

In addition to the standard layers provided with EPLAN, you can, of course, create your own layers. You should be somewhat systematic in doing so, otherwise you will end up with “proliferated chaos”. Layers can only be (directly) created in existing nodes such as graphics or property placements. New (superior) nodes cannot be directly created.

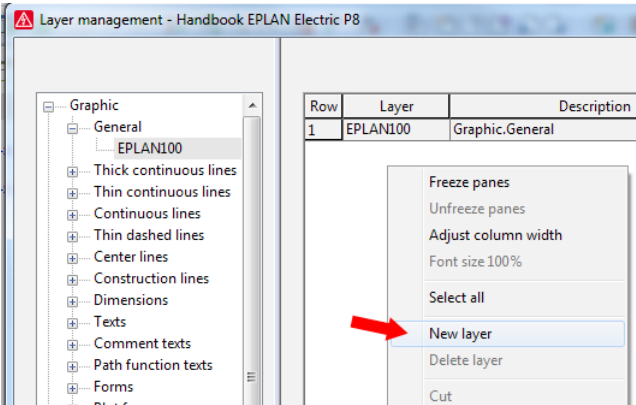


Fig. 7.66 Creating a new layer

You create new layers from the **LAYER MANAGEMENT** dialog by right-clicking and selecting the **NEW LAYER** popup menu item (in the right area of the **LAYER MANAGEMENT** dialog), or by using the  button. Use the mouse to select the node in which the desired layer should be created. Now, call the *New layer* function. EPLAN generates a new layer with the name **NEW_LAYER_1**.

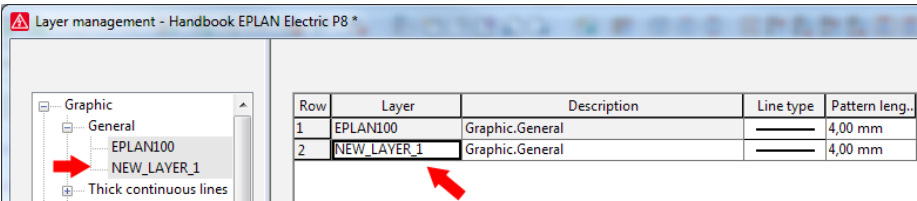


Fig. 7.67 New user-created layer

This layer can now be edited. The key columns here are **Layer** and **Description**.

The **Layer** column bears the name of the layer. When you create your own layers, you should give them sensible names. One possibility is to name the layers after the company and to include a range of numbers, e.g. COMPANY1000.

The **Description** column describes the layer itself. This description can be viewed later at the time of assigning layers to individual objects in the project. You should therefore enter a descriptive text here that allows a function to be assigned to the layer based on the description.



The layer will be later sorted into the left-hand area of layer management on the basis of the description.

We will now change the layer from NEW_LAYER_1 to HBE-0100. The layer description will be changed to **Handbook.Graphic.Special**. Immediately after you save the layer, EPLAN will sort it as follows.

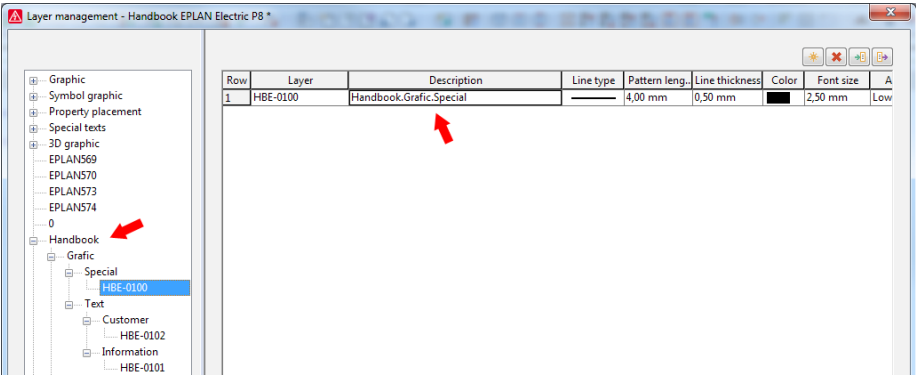


Fig. 7.68 Newly sorted layer

Properties such as *Line type*, *Pattern length*, *Line thickness*, etc. are self-explanatory and are stored or selected in accordance with the layer requirements. It is also possible to define default values for angles and their orientation, whether this layer should be in the background, or whether the layer and its elements should be scalable.

Properties of layers

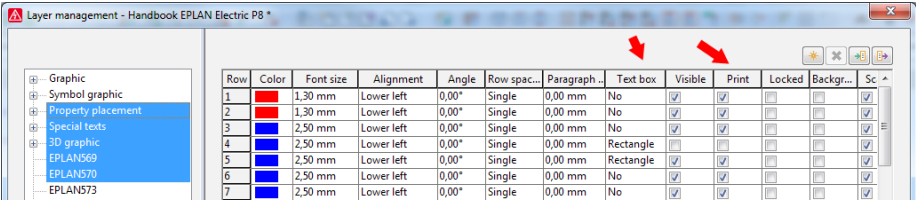


Fig. 7.69 Other layer settings options

7.3.4 Uses of layers

What can you use all these layers for? You can use your own layers, for example, to provide extra project information for the workshop. This could be cabling notes or deadlines. The possible uses are limitless.



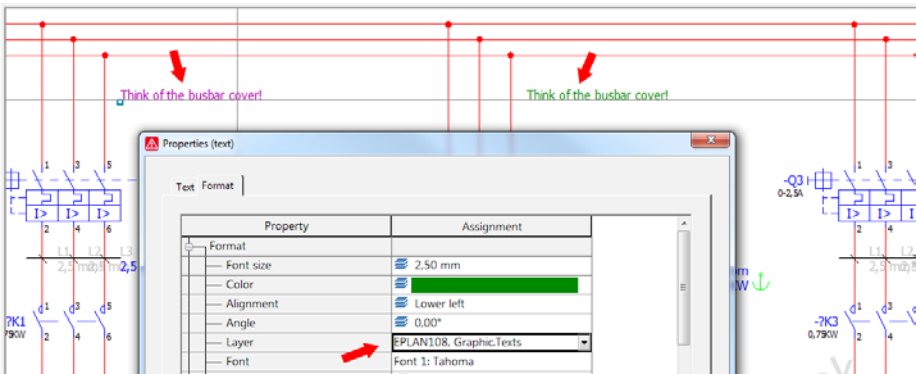
As described in section 7.3.3, a new layer HBE-0101 with the description **Handbook.Graphic.Texts.Workshop.Information** description was created. This layer is for displaying further information for workshop production in the schematic. When the workshop has finished using this information, it should then be easy to hide this information for the entire project.

Fig. 7.70
Workshop layer

Row	Layer	Description	Line type	Pattern leng.	Line thickness	Color	Font size	A
1	HBE-0100	Handbook.Grafic.Special		4,00 mm	0,50 mm		2,50 mm	Low
2	HBE-0101	Handbook.Grafic.Text.Information		4,00 mm	0,35 mm		3,50 mm	Low
3	HBE-0102	Handbook.Grafic.Text.Customer		4,00 mm	0,35 mm		2,50 mm	Low
4	HBE-0400	Handbook.Eigenschaftsplatzierung.Kabel.D...		4,00 mm	0,35 mm		2,50 mm	Low

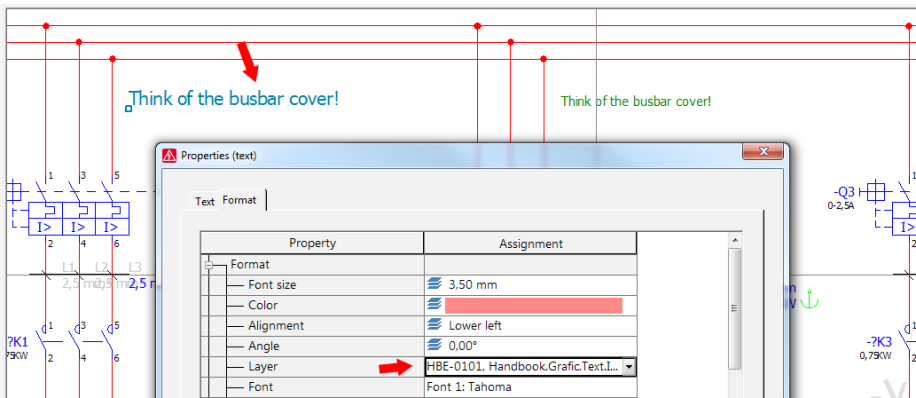
A schematic page with corresponding entries is then opened in the project. Now, special information for the workshop is placed on the page as text using the EPLAN default settings for texts.

Fig. 7.71
Text with EPLAN
default setting



Then, these texts are selected and placed on the newly created layer **Handbook.Graphic.Text.Information**.

Fig. 7.72
Texts placed on the
new layer



EPLAN handles the settings that pertain to this layer and automatically formats the texts accordingly. Via layer management, this workshop information can now be hidden, for example, prior to printing.

Row	Layer	Angle	Row spac...	Paragraph ..	Text box	Visible	Print	Locked	Backgr...	Scalable	3D layer
1	HBE-0100	0,00°	Single	0,00 mm	No	☒	☒	☐	☐	☐	☐
2	HBE-0101	0,00°	Single	0,00 mm	No	☒	☒	☐	☐	☒	☐
3	HBE-0102	0,00°	Single	0,00 mm	No	☒	☒	☐	☐	☒	☐
4	HBE-0400	0,00°	Single	0,00 mm	No	☒	☒	☐	☐	☒	☐

Fig. 7.73 Excluding a layer from printing

This is only a small example of what is possible with layer management. The user has practically unlimited possibilities here. The biggest advantage of layers is that it is easy to globally change the format, representation, etc. of the respective element on this layer with just a few mouse clicks.

7.3.4.1 Editable layer properties

EPLAN provides the following properties/selections for layers, which can be changed globally in the layer management.

Property	Selection or input	Comment
Line type	Drop-down list	51 line types available
Pattern length	Free input	–
Line thickness	Free input	–
Color	Selection	Branching to color selection
Font size	Free input	–
Alignment	Drop-down list	Selection of 12 values
Angle	Free input	–
Row spacing	Drop-down list	Selection of 3 values
Paragraph spacing	Free input	–
Text box	Drop-down list	Selection of 4 values
Visible	Check box	Yes/No
Printing	Check box	Yes/No
Locked	Check box	Yes/No
Background	Check box	Yes/No
Scalable	Check box	Yes/No
3D layer	Check box	Yes/No

7.3.4.2 Remove unnecessary layers from the project

To remove layers that have been rendered superfluous, for example, layers that were stored in the project and subsequently changed as a result of copy actions or the insertion of DXF/DWG files, you can use the compress the project.

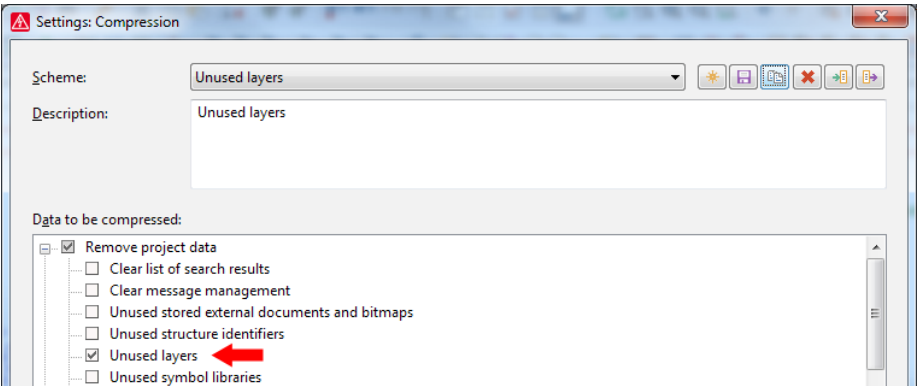


Fig. 7.74 Compression scheme to remove unnecessary layers

After the scheme is launched, EPLAN removes the unnecessary layers and then displays a summary of these removed layers.

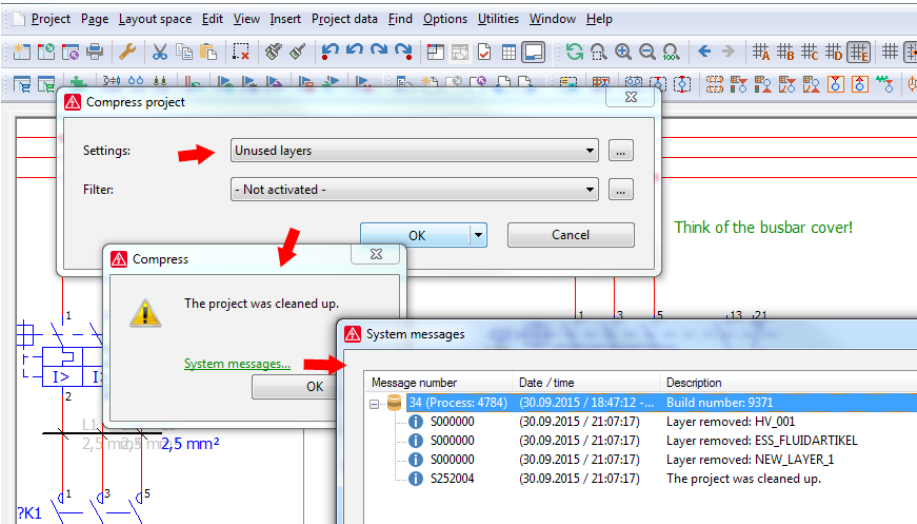


Fig. 7.75 System message

7.4 Parts management

All devices in EPLAN, with all their technical and commercial data such as technical characteristics, dimensions (width, height, depth) or prices, are managed in **Parts management**. However, not only device-specific data is managed in parts management – the corresponding function definitions, symbols and also symbol macros can be stored for every device.

Parts management is called via the **UTILITIES > PART > MANAGEMENT** menu.

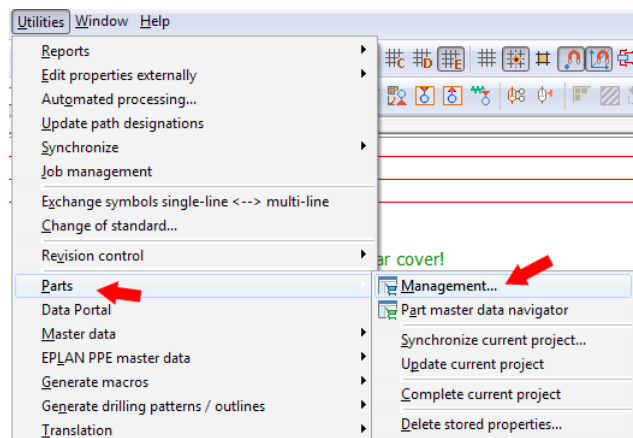


Fig. 7.76
Calling up parts management

The actual parts database used is defined via **OPTIONS/SETTINGS/USER/MANAGEMENT/PARTS MANAGEMENT**. You can also run parts management from an SQL server database instead of an Access database.

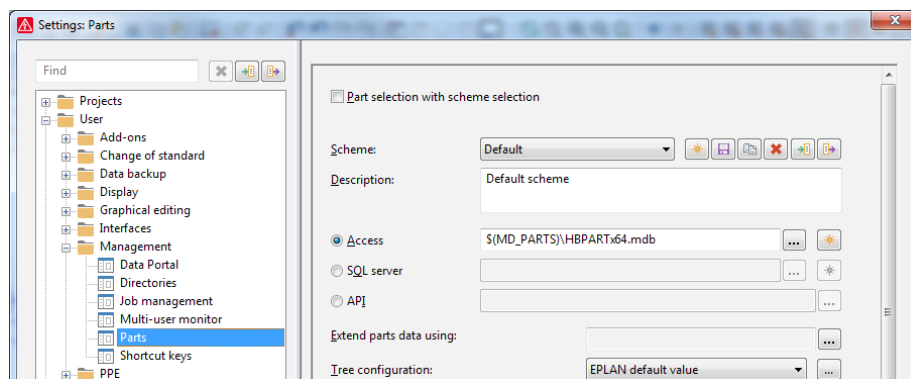


Fig. 7.77
Parts management user settings

In these **User-specific settings**, you can also set the parts management **View** (tree configuration). There are various options to adjust the *tree configuration* view to your own specifications by using your own schemes.

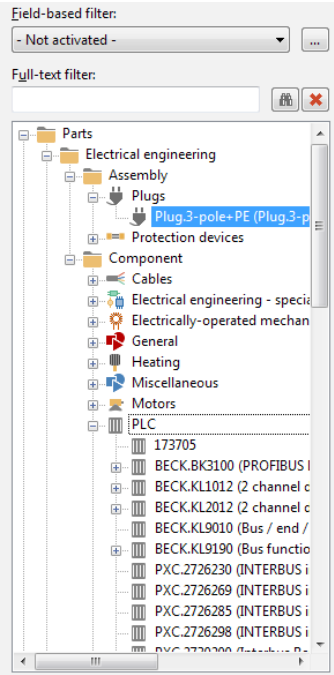


Fig. 7.78 Standard EPLAN view

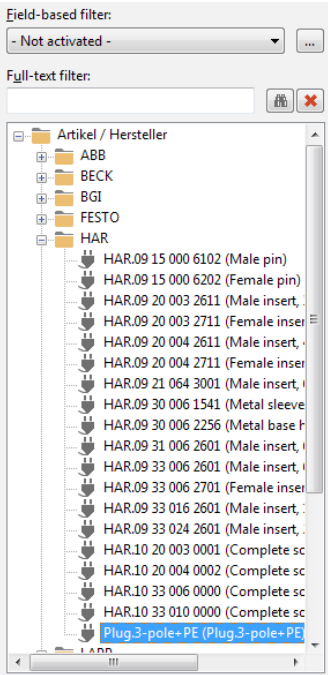


Fig. 7.79 View by manufacturer

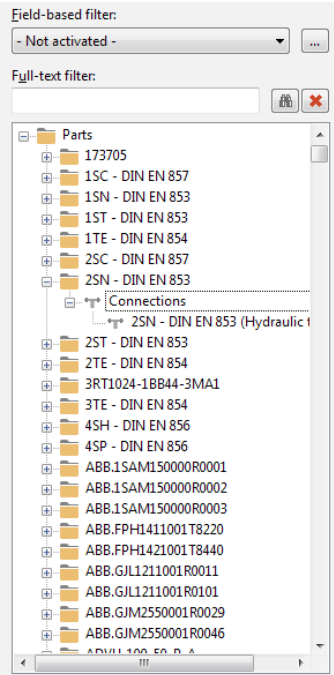


Fig. 7.80 View by manufacturer

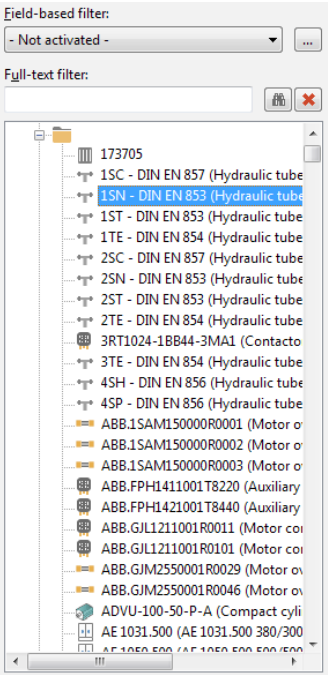


Fig. 7.81 View by ERP number

7.4.1 Structure of parts management

It would take an entire book to describe parts management in detail. In this section, I cover the most important information that is of interest and useful in everyday practice. More information on this topic can be found in my book *EPLAN Electric P8 Artikelverwaltung* (ISBN 978-3-446-43332-8, only available in German).

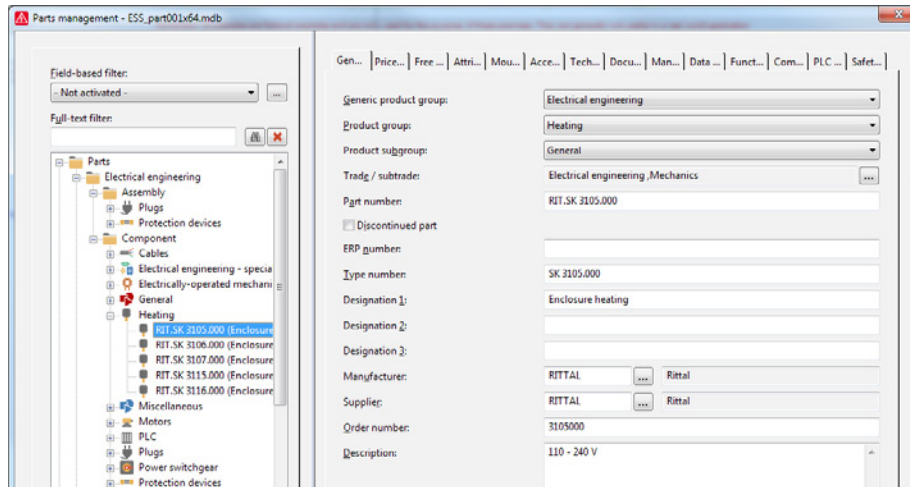


Fig. 7.82 Overview of the parts management dialog

The PARTS MANAGEMENT dialog basically consists of an area on the left containing an overview of all parts, customers and manufacturers/suppliers – and an area on the right containing detailed information. Both parts are separated by “splitters” and, therefore, scalable to (almost) any size. The area on the right always relates to one (or more) selected parts in the area on the left – the parts or customers and/or manufacturers/suppliers. The area on the right, i.e. the parts data, is further divided into several tabs containing the various data relating to the part.

7.4.2 Tabs in parts management

Each part has several tabs. Various technical and commercial data for the part are defined in these tabs. The existing tabs of the part Siemens SIE.3LD2 504–0TK53 will serve as our example here.

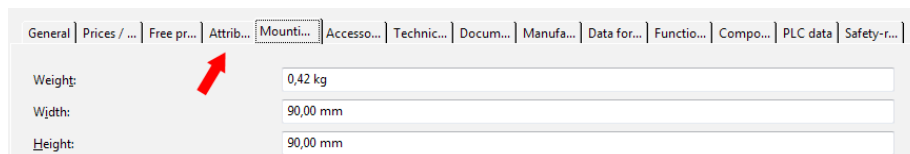


Fig. 7.83 The different tabs of a part



NOTE: The tabs shown here may differ from part to part. For example, there are differences for terminals, plugs, PLC, etc. You need to take this into consideration when editing.

The screenshot shows the 'General' tab of a part management dialog. The tabs at the top are: Gen..., Pric..., Fre..., Attr..., Mo..., Acc..., Tec..., Doc..., Ma..., Dat..., Fun..., Co..., PLC..., Safe... The 'General' tab is active. The fields are as follows:

- Generic product group:** Electrical engineering (dropdown, highlighted with a red arrow)
- Product group:** Power switchgear (dropdown)
- Product subgroup:** Undefined (dropdown)
- Trade / subtrade:** Electrical engineering (dropdown with ellipsis)
- Part number:** SIE.3LD2 504-0TK53
- ☐ Discontinued part
- ERP number:** (empty field)
- Type number:** 3LD2 504-0TK53
- Designation 1:** Main / Emergency stop switch
- Designation 2:** (empty field)
- Designation 3:** (empty field)
- Manufacturer:** SIEMEN (dropdown with ellipsis) Siemens
- Supplier:** SIEMEN (dropdown with ellipsis) Siemens
- Order number:** 3LD2 504-0TK53
- Description:** 3-pole
IU=63
P/AC-23 A at 400 V=22 kW
Front mounting
Four-hole mounting Rotary actuator red/yellow

Fig. 7.84 General tab

The *General* tab contains general information on the part, such as the part number, ERP number, **product group** (e.g. a contactor or the **trade**). You will also find information here such as the **part number**, **ERP number**, **order number**, the device **designations** and the **manufacturer**.



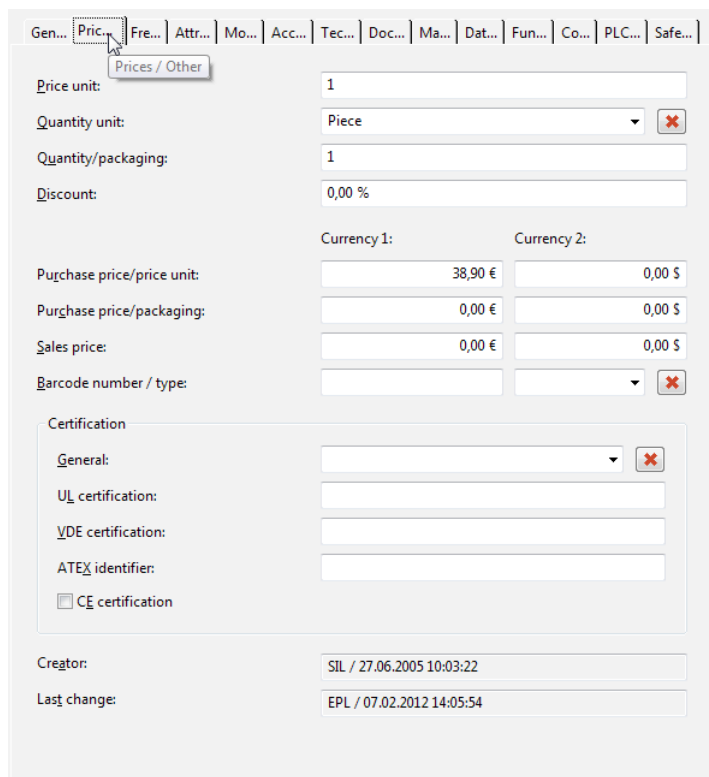
NOTE: A part is identified by its part number and the ERP number. This means that they have to be unambiguous and can only be used once in parts management.

Most of these fields are pure input fields; where this is impossible, for example for the generic product group, there are fixed selection lists. If a record for the manufacturer and/or supplier has been created in parts management, the *Manufacturer* and *Supplier*

fields can be selected from the **MANUFACTURER** and/or **SUPPLIER DIALOG** that opens when you click . EPLAN then applies this entry and automatically fills in the full name of the manufacturer or supplier.



NOTE: The user cannot add entries to these selection lists (*generic product group, product group and product subgroup*). Only EPLAN can do that.



Gen... | **Pric...** | Fre... | Attr... | Mo... | Acc... | Tec... | Doc... | Ma... | Dat... | Fun... | Co... | PLC... | Safe...

Prices / Other

Price unit: 1

Quantity unit: Piece 

Quantity/packaging: 1

Discount: 0,00 %

Currency 1: Currency 2:

Purchase price/price unit: 38,90 € 0,00 \$

Purchase price/packaging: 0,00 € 0,00 \$

Sales price: 0,00 € 0,00 \$

Barcode number / type: 

Certification

General: 

UL certification:

VDE certification:

ATEX identifier:

☐ CE certification

Creator: SIL / 27.06.2005 10:03:22

Last change: EPL / 07.02.2012 14:05:54

Fig. 7.85 Prices/Other tab

The *Prices/Other* tab contains commercial and organizational entries like *Quantity unit*, *Package sizes*, *Prices* and even linked documents for *Certification*.

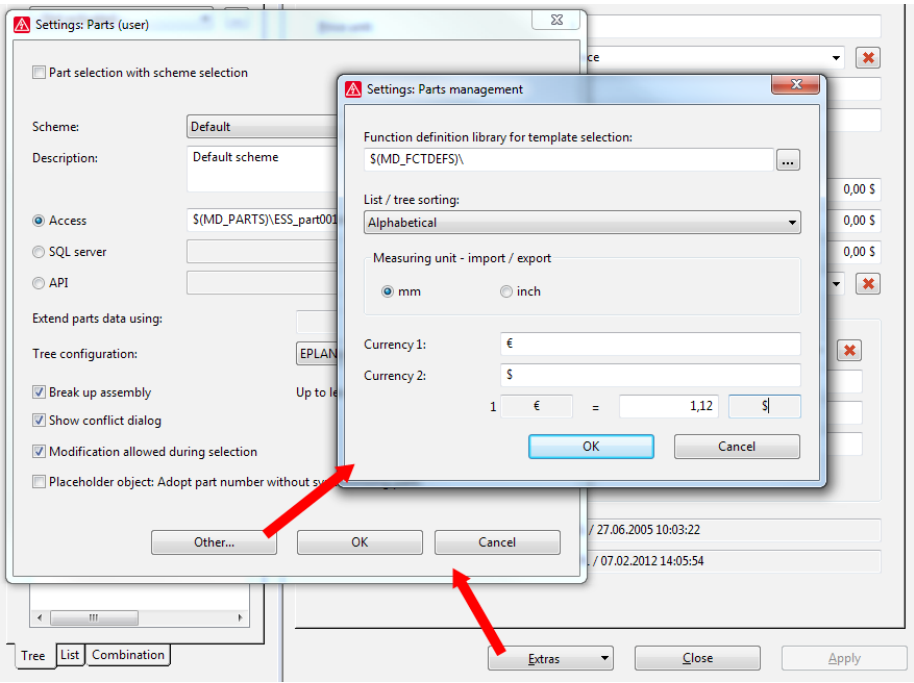


Fig. 7.86 Selecting a currency

To define the currency, you first click the **EXTRAS** button in parts management, then select the **SETTINGS** menu item, and then click the **MORE** button in the **SETTINGS/PARTS (USER)** dialog.

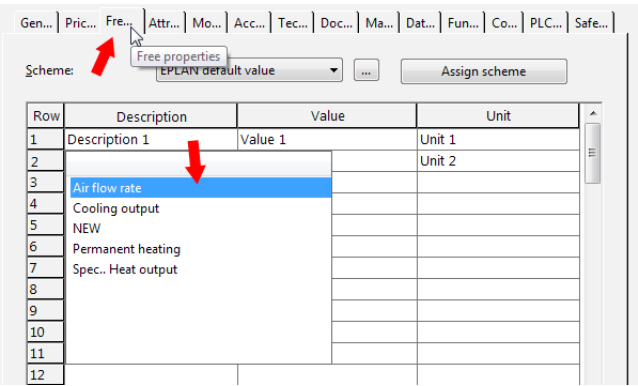


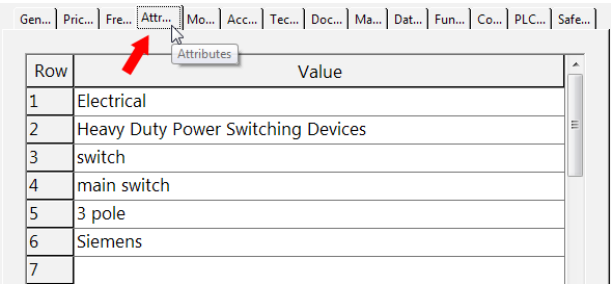
Fig. 7.87 Free properties tab

The *Free properties* tab covers 100 properties that are freely available for any desired purpose. Each of these free properties consists of a *Property*, a *Value* for the property, and a *Unit*.

All fields can be freely used, for instance, for information that is not available as a standard field for the part. Block properties can be used with the free properties to display necessary or additional information at a specific symbol in the schematic (with stored parts).

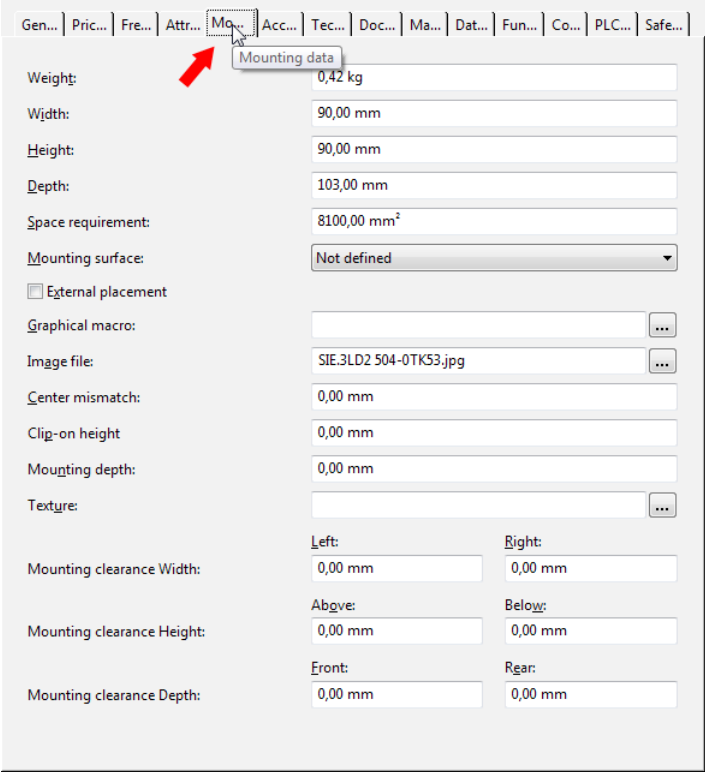
You can create your own schemes with free assignment of the free properties and assign parts to these schemes without having to manually enter the contents of the free properties for every part.

The *Attributes* tab allows you to store additional information (in the *Value* column) that is not available as standard in parts management or used elsewhere. For example, these entries can be used for filtering or as a sorting characteristic for the tree configuration. The entries themselves are limited to 200 characters.



Row	Value
1	Electrical
2	Heavy Duty Power Switching Devices
3	switch
4	main switch
5	3 pole
6	Siemens
7	

Fig. 7.88
Attributes tab



Weight:	0,42 kg
Width:	90,00 mm
Height:	90,00 mm
Depth:	103,00 mm
Space requirement:	8100,00 mm ²
Mounting surface:	Not defined
☐ External placement	
Graphical macro:	
Image file:	SIE.3LD2 504-0TK53.jpg
Center mismatch:	0,00 mm
Clig-on height	0,00 mm
Mounting depth:	0,00 mm
Texture:	
Mounting clearance Width:	Left: 0,00 mm Right: 0,00 mm
Mounting clearance Height:	Abgve: 0,00 mm Below: 0,00 mm
Mounting clearance Depth:	Front: 0,00 mm Rear: 0,00 mm

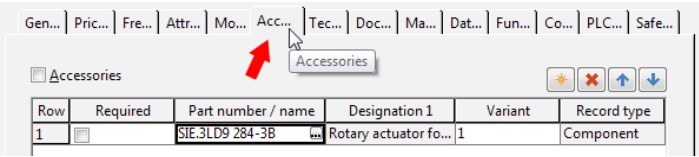
Fig. 7.89
Mounting data tab

The *Mounting data* tab contains all the information that is necessary to use the part, for example, for a mounting panel layout. It is recommended to meticulously maintain this information, such as *Width*, *Height* or *Depth* (especially important for the 3D mounting layout (EPLAN Pro Panel)) for the part and to update any missing data. A stored image file can also be useful, because this information is evaluated at different locations in EPLAN. This greatly simplifies work steps such as equipping mounting panels or generating data reports like parts lists, device tag lists, etc. because the necessary data can be processed automatically from parts management.



NOTE: The *Graphic macro* field on this tab is intended for the Pro Panel (3D macro). For 2D panel layout, the 2D macro should be stored in the *Macro* field on the *Technical data* tab.

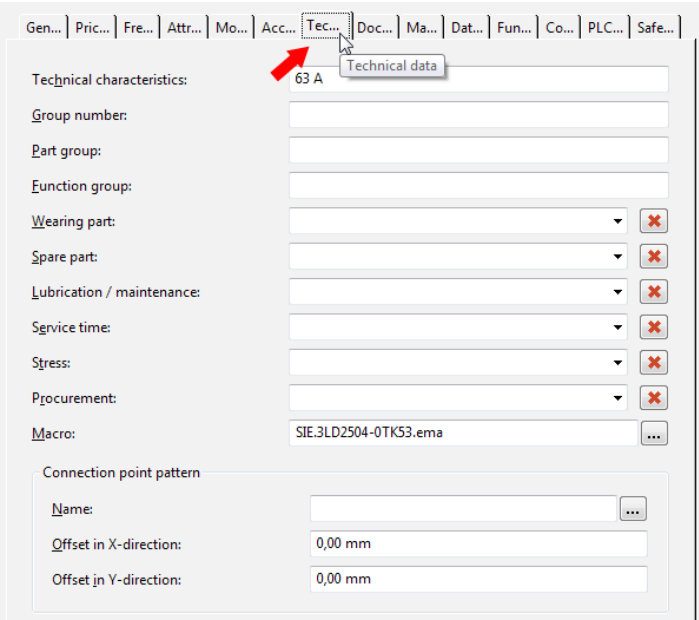
Fig. 7.90
Accessories tab



The *Accessories* tab allows you to assign specific accessories to “main” parts. This accessory is also created as a normal part. You turn it into an accessory part by activating the *Accessory* check box. Then it can be selected and assigned under the “main” parts on the *Accessory* tab.

Selected accessories can be marked as **required** or, if the identifier is missing, as optional accessories.

Fig. 7.91
Technical data tab



The *Technical data* tab contains part data such as *service time* and whether the part is a *wearing part*. Most fields are pure input fields, such as the *Group number* or *Function group* fields.

The *Wearing part*, *Spare part*, *Lubrication/maintenance*, *Service time*, *Stress* and *Procurement* fields are also input fields. However, each entry is recognized by EPLAN and written to a selection list (internal) so that, over time, this selection list contains many entries that have already been used (and can be selected again).

The remaining fields of this tab can be selected via the  selection button. The *Macro* field can contain the corresponding window macro for the project. Using the familiar macro functions with the different representation types (e.g. part placement, overview or other representation types), this macro can be set up in such a way that it can be put to multiple uses during project editing.

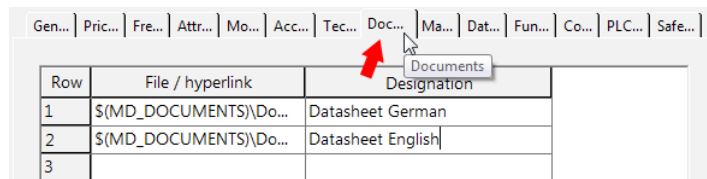


Fig. 7.92 Documents tab

The *Documents* tab can hold up to 20 different documents. These can be user instructions, manuals, etc. Different language versions of the documents can also be stored for the part. The value in the *Description* can be in multiple languages. This means that the field can be translated.

The *Manufacturing* tab contains a relevant drilling pattern for the selected part. The procedure here is similar to that for accessories lists. Here, too, the drilling pattern must be available before the actual assignment, that is, before it is assigned to the part.

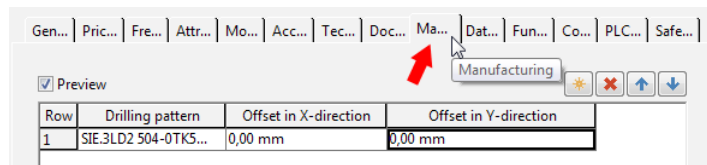


Fig. 7.93 Manufacturing tab

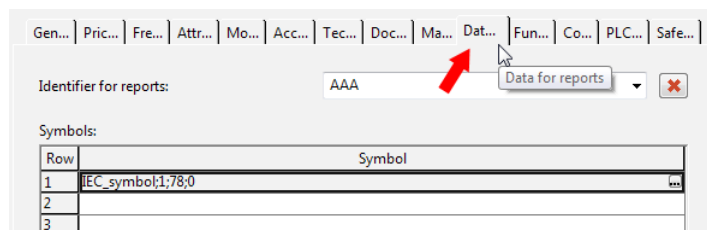


Fig. 7.94 Data for reports tab

It makes sense to use the *Data for reports* tab in combination with the **device tag list** report type and conditional forms (so-called subforms for dynamic forms). This way, certain graphical and more complex requirements regarding the output of parts in a device tag list can be displayed in a specific manner.

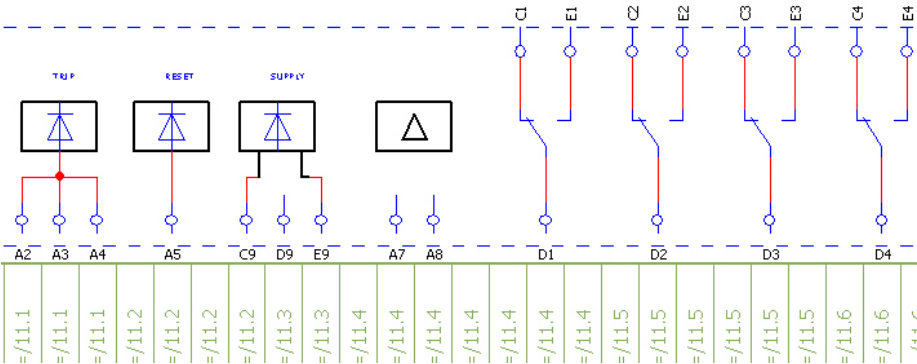


Fig. 7.95 Example view of a complex graphical report of a part

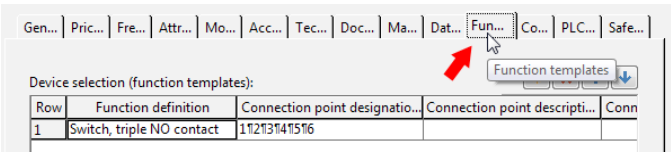


Fig. 7.96 Function templates tab

The *Function templates* tab contains key information about the actual part and for the subsequent device selection for the part. The *Function templates* for selecting devices for a part are defined in this tab. I.e. this is where you define whether this part is a lamp or an auxiliary contact.

EPLAN needs function templates in order to compare the devices already used in the project with the devices in parts management when performing device selection (not to be confused with parts selection). This allows you to select devices that match the items in the schematic.

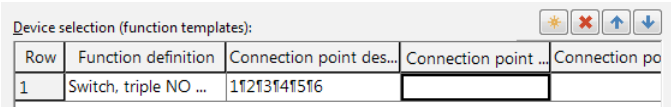


Fig. 7.97 The first columns of the function template

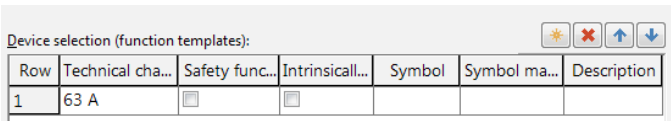
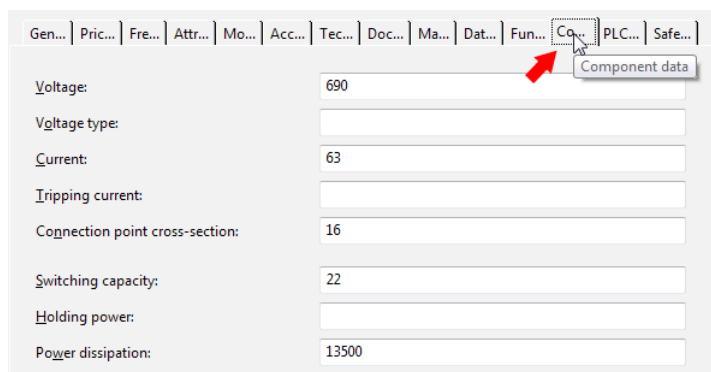


Fig. 7.98 Further function template columns

Entries such as the **Function definition**, the **Connection point designation**, the **Connection point description** or the **Characteristic** allow comparison criteria to be defined. For example, when selecting a device, a pushbutton that was defined in parts management with a pushbutton function definition and connection point designations 7 and 8 can only be assigned to an item in the schematic that has exactly these properties, assuming that the device selection settings have been set for this.

For each function definition, you can define, in addition to the previous values, specific **symbols** or **symbol macros** that EPLAN is then to use for graphical project editing should the standard symbol not fit. You can also define for each function definition whether the function is *relevant to safety* or *intrinsically safe*.



Field	Value
Voltage:	690
Voltage type:	
Current:	63
Tripping current:	
Connection point cross-section:	16
Switching capacity:	22
Holding power:	
Power dissipation:	13500

Fig. 7.99 Component data tab

The *Component data* tab contains further technical data for the part, which can also be evaluated – for use in reports, calculations or simply as an additional representation in the schematic on the device.

The second-to-last tab, *PLC data*, contains information that is important for PLC devices. Here you can enter PLC type designations, enter object descriptions, or generate reports for them.

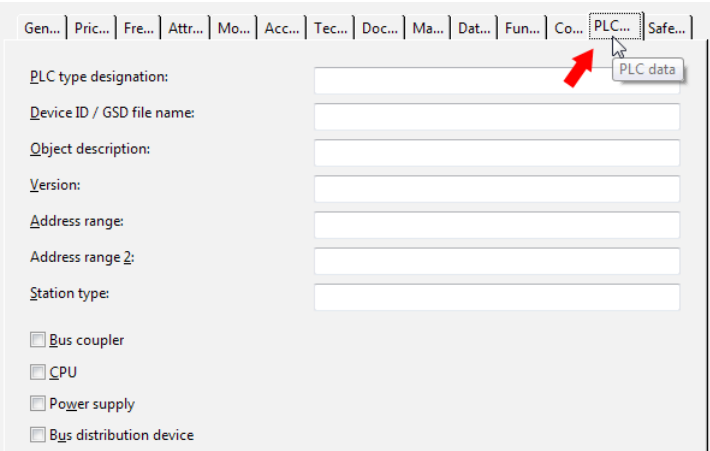


Fig. 7.100 PLC data tab

The last tab, *Safety-related values*, contains important information on a device's characteristic values (performance level, SIL, etc.), use cases, and much more.

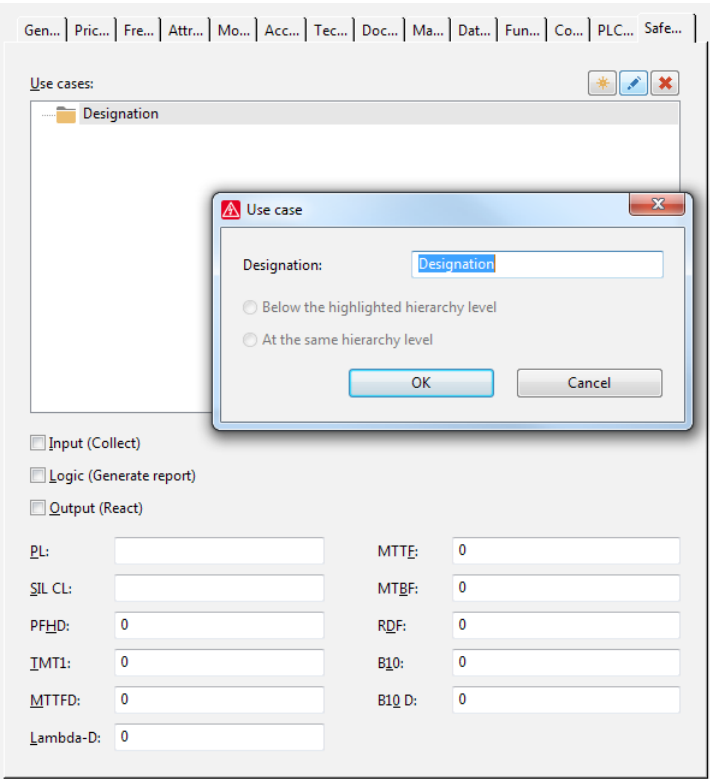


Fig. 7.101 Safety-related values tab

■ 7.5 Revision control

In addition to actual project creation, EPLAN Electric P8 provides a management tool that allows you to record, evaluate or document revisions (e.g. changes to the site, end customers, etc.) to this project at the end of project editing.

In EPLAN Electric P8, the task of recording, managing and documenting revisions is handled by revision control. EPLAN distinguishes between two fundamental procedures for executing revision control: Revision based on a **project comparison** and revision based on **change tracking**.

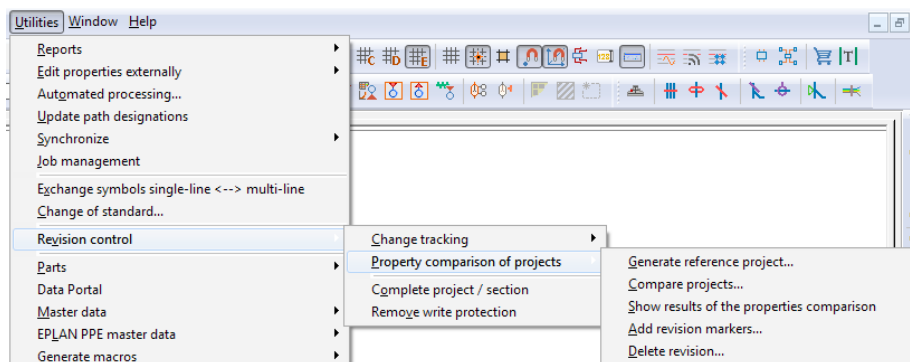


Fig. 7.102 Revision control with property comparison

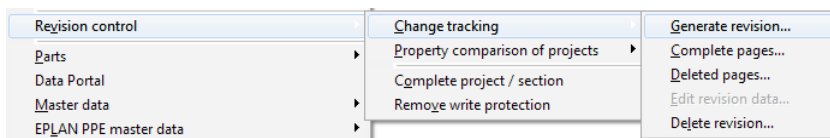


Fig. 7.103 Revision control with change tracking

7.5.1 General

A revision in EPLAN Electric P8 is essentially defined as follows: A revision is an edited project in which the various changes are marked and then tracked via revision. A revised project is stored with the file extension **.ell*.

The following will explain revision control using change tracking as an example.

7.5.2 Generate new revision

Before a revision, and thus change tracking of modified data, can be executed, you must first activate it (the revision) for the project. In the UTILITIES/REVISION CONTROL/CHANGE TRACKING menu, start the GENERATE REVISION menu item.

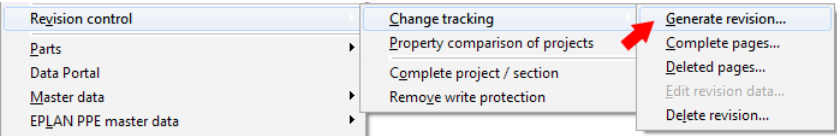


Fig. 7.104 Generating revision

EPLAN opens the **GENERATE REVISION** dialog. Here you must enter a *Revision name*. The *Comment* on the revision is optional. The *user*, *defined working section*, and the *date* cannot be changed and are set by EPLAN automatically. This data always applies to a complete revision. It is a kind of global “header” for the revision data. This data in the **GENERATE REVISION** dialog is part of the project properties.

Fig. 7.105
Entering revision data

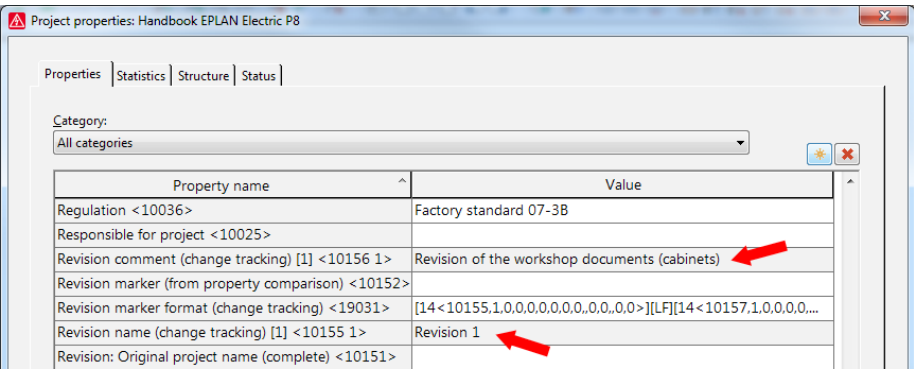
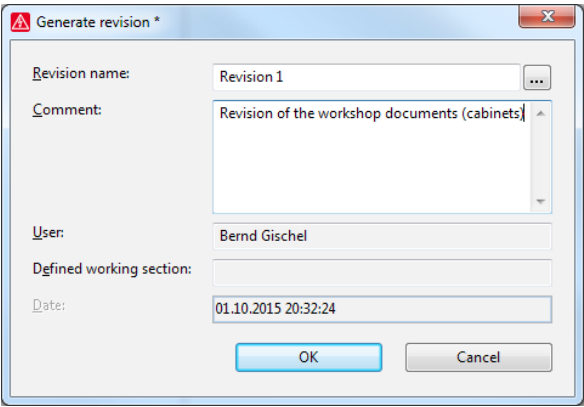


Fig. 7.106 Project property names of revision entries

This global revision data can, for example, be integrated into the plot frame. If all data has been entered, you can close the **GENERATE REVISION** dialog by clicking OK. EPLAN now generates the revision and, for the purposes of the visual check, changes the project icon in the page navigator.

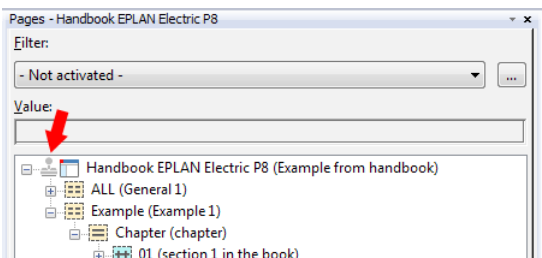


Fig. 7.107 Modified icon

7.5.3 Execute changes

After the revision has been created, the changes can be entered in the schematics.

If EPLAN is to query the index automatically with every change, you must set *Always prompt for description of page modification* in the settings (**OPTIONS/SETTINGS/PROJECTS/[PROJECT NAME]/MANAGEMENT/REVISION (CHANGE TRACKING)** menu).

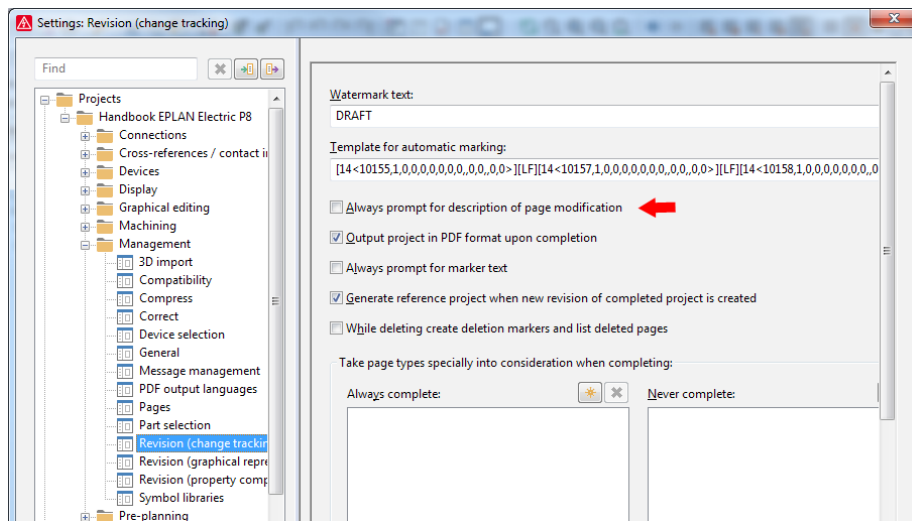


Fig. 7.108 Always prompt for page modification setting

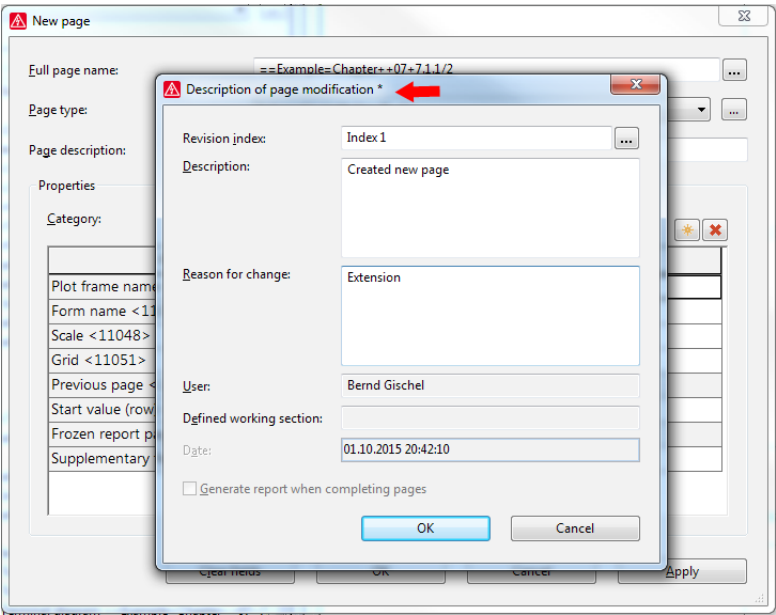


Fig. 7.109 Index query following a change

After each change, EPLAN prompts for a revision index. EPLAN opens the **DESCRIPTION OF PAGE MODIFICATION** dialog. Here the *Revision index* is defined as well as a *Description*, which can also be optional. The *user*, *defined working section*, and the *change date* are also pre-set by EPLAN from the system.

For the *revision index*, you should choose one system to avoid chaos and to prevent the numbering of the revision index from becoming mixed up upon completion of the project. It would be a good idea to integrate the revision index, and possibly its description, into the plot frame. You would immediately be able to see, on a printout for example, that a page has been modified.



Index 1	01.10.2015	Bernd Gischel	Date	01.10.2015
			Ed. by	Bernd Gischel
			Appr	Bernd Gischel
Modification	Date	Name	Original	

Fig. 7.110 Adding the revision index to the plot frame

The properties of the revision index and their additional data such as the description are pure page properties.

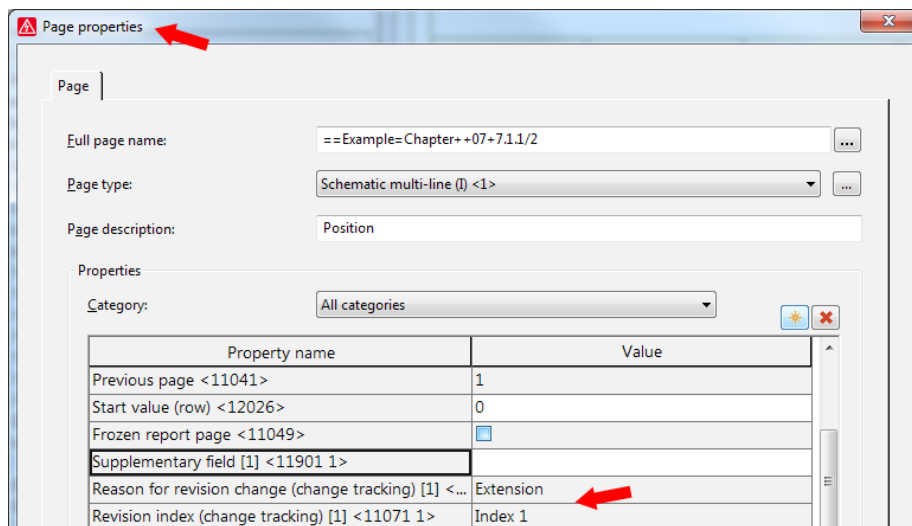


Fig. 7.111 Revision index in the page properties

Generally, you can now make any desired modifications to the project. Changes on a page are marked with the **Watermark** (the text of the watermark can be modified in the settings) across the plot frame. This serves as a visual indication that this page and the modifications have not been completed yet.

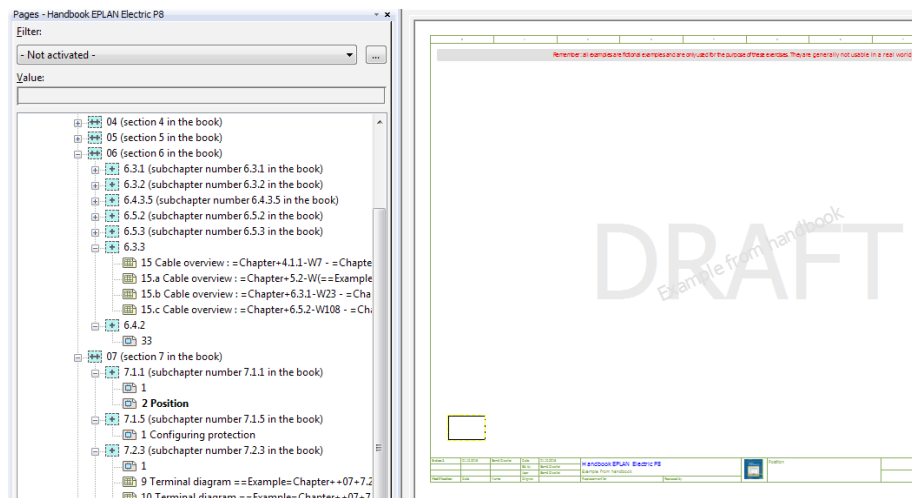


Fig. 7.112 Marking the page as DRAFT

EPLAN allows you to place revision markers for every modification. You can also use visually different revision markers for different types of changes.

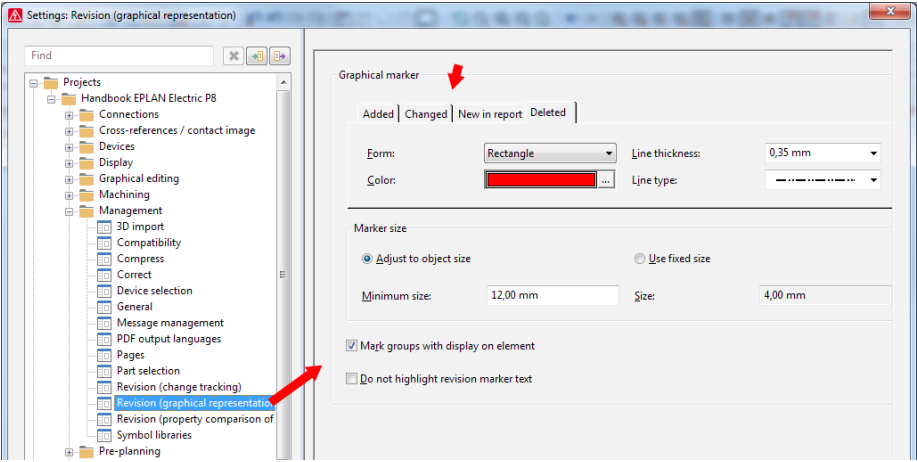


Fig. 7.113 Graphical markers for various modifications

Here, EPLAN differentiates between *Added*, *Changed*, *New in report* and *Deleted*. Depending on your requirements, you can adjust these graphical markers for each project differently.

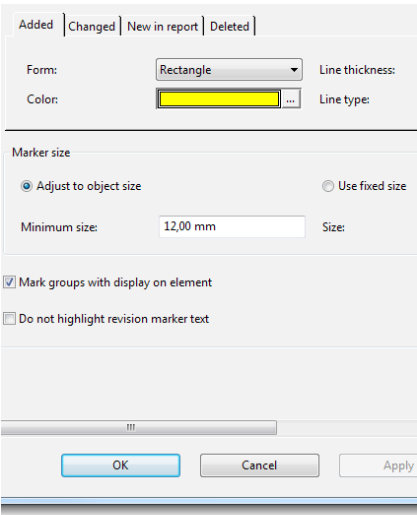


Fig. 7.114 Graphical marker for added objects

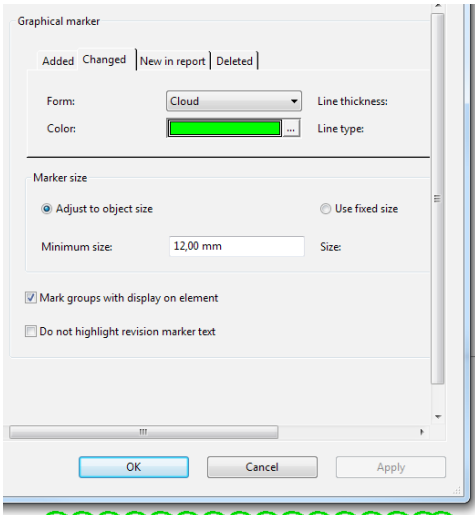


Fig. 7.115 Graphical marker for changed objects

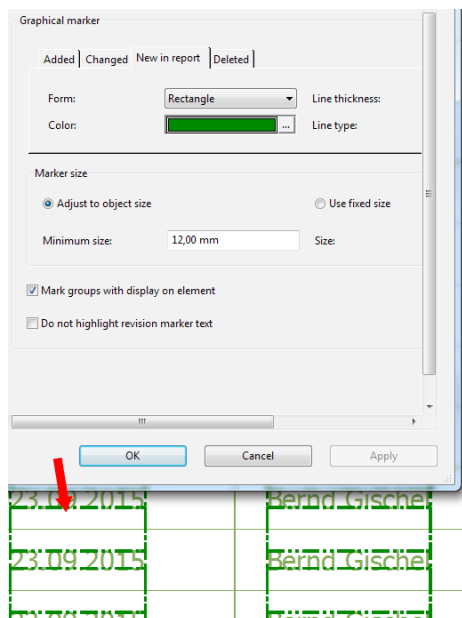


Fig. 7.116 Graphical marker for “New in report”

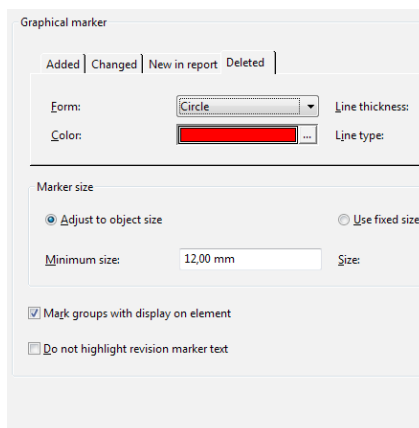


Fig. 7.117 Graphical marker for deleted objects

7.5.4 Complete page(s)

After the modifications have been incorporated into the project, you need to complete the page or pages. Go to the UTILITIES/REVISION CONTROL/CHANGE TRACKING menu and start the COMPLETE PAGES menu item.

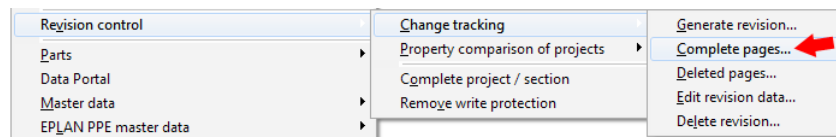


Fig. 7.118 Complete pages

EPLAN opens the **DESCRIPTION OF PAGE MODIFICATION** dialog. Here you must now enter the *revision index*. It is also possible to switch to the **SELECT REVISION INDEX** dialog via the More button. From there, select, apply or modify an existing revision index.

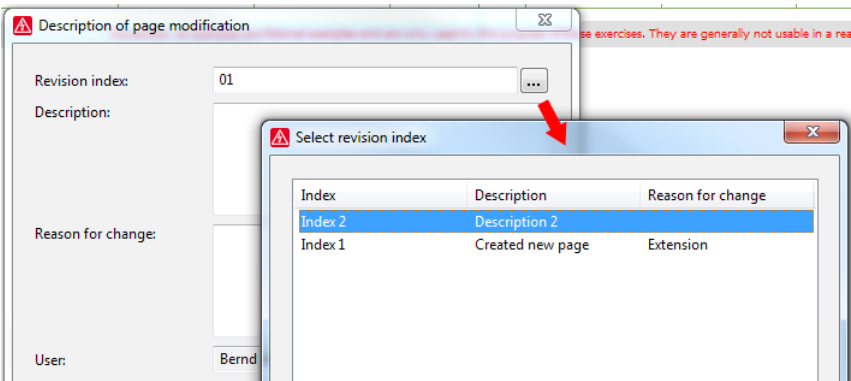


Fig. 7.119 Defining the description of page modification

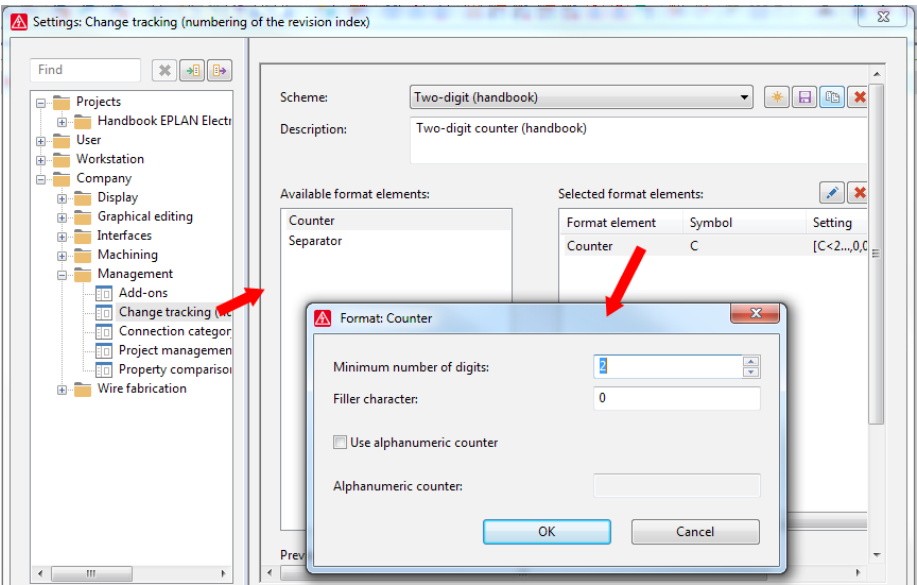


Fig. 7.120 Defining the numbering of the revision index

If a numbering scheme has been defined for the revision index in the settings (OPTIONS/UTILITIES/COMPANY/MANAGEMENT/CHANGE TRACKING (NUMBERING SCHEME FOR REVISION INDEX)), a revision index will be recommended automatically based on the selected scheme.

NOTE: As long as the project itself has not been completed, any completed page can be modified again. In that case, you merely have to complete the page again.

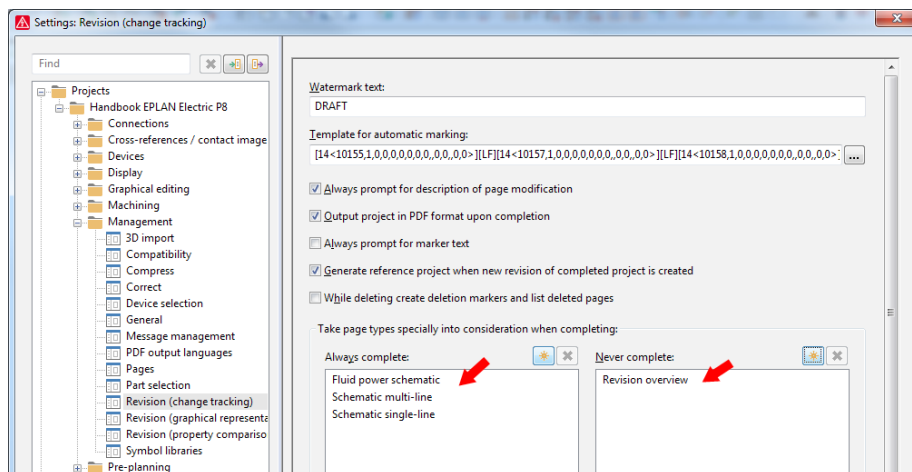


Fig. 7.121 Special handling of page types

The **Always complete pages** option means that these pages are always assigned a revision index even if no modifications have been made. But EPLAN can assign an automatic revision index only if a scheme for this has been defined in the settings.

Never complete pages has the opposite effect. These pages are ignored for revisions even if modifications have been made.

7.5.5 Generate reports

Once all modifications have been made, you can generate and/or update the reports. Special revision overviews will give you an overview of the changes made to the project.

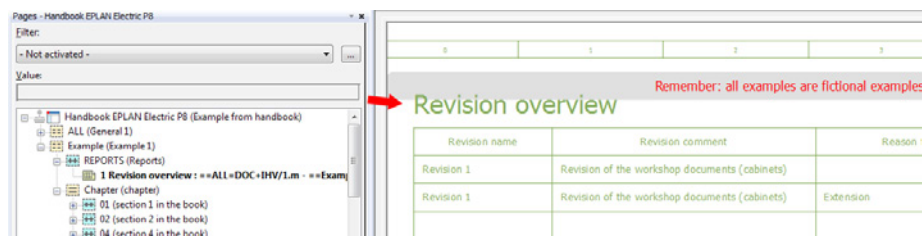


Fig. 7.122 Revision overview report

A special menu item allows you to list even the pages that were deleted from the project. You can call up this menu item via **UTILITIES/REVISION CONTROL/CHANGE TRACKING/DELETED PAGES**. When this function is started, EPLAN displays the **DELETED PAGES** dialog. This contains information about when and how pages have been deleted (e.g. when reports were updated), plus the data you entered, such as a reason for the change.

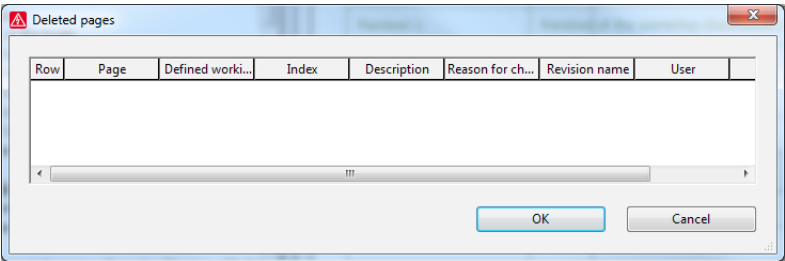


Fig. 7.123 Deleted pages dialog

7.5.6 Complete a project

If all modifications have been finished, the reports generated and updated, and the pages completed, you are ready to complete the project itself. Go to the **UTILITIES/REVISION CONTROL** menu and start the **COMPLETE PROJECT/AREA** menu item.

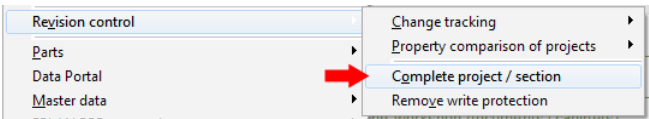


Fig. 7.124 Complete project/area

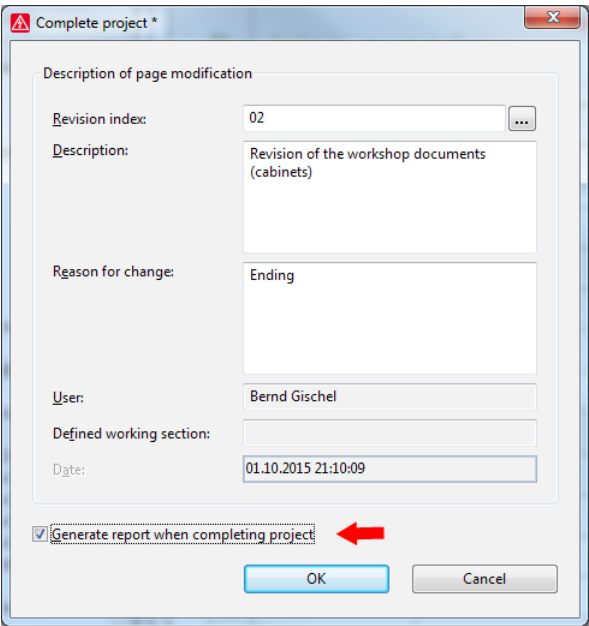


Fig. 7.125 Complete project

EPLAN opens the **COMPLETE PROJECT** dialog. Here, you enter the already familiar data like revision index or the reason for the change. If you select the **Generate report when completing project** option, the following happens: EPLAN updates the project, evaluates all reports and completes each page, i.e. it removes the entry *Draft* and/or the watermark. All revision indices that EPLAN assigns for this automatically now show the most recently defined revision index.

0	1	2	3
Revision overview			
Remember: all examples are fictional examples			
Revision name	Revision comment	Revision index	
Revision 1	Revision of the workshop documents (cabinets)	Ending	
Revision 1	Revision of the workshop documents (cabinets)	Ending	
Revision 1	Revision of the workshop documents (cabinets)	Ending	
Revision 1	Revision of the workshop documents (cabinets)	Ending	

Fig. 7.126 Automatically completed project

A completed project is represented in the page navigator by a corresponding icon.

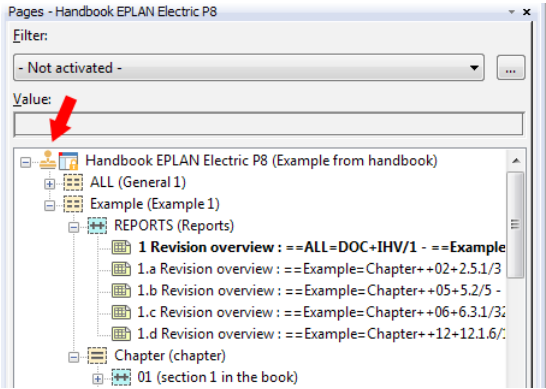


Fig. 7.127 Icon for a completed project

7.6 Project management

The previous sections already mentioned in part the EPLAN project management. This section will help gain a deeper understanding of project management, as well as show further options and functions in EPLAN.

7.6.1 Project management dialog

Project is opened via the **PROJECT/MANAGEMENT** menu item.

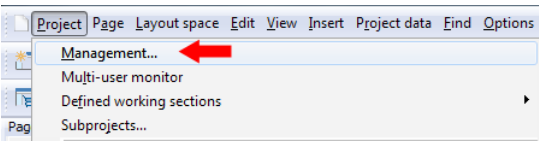


Fig. 7.128 Opening project management

EPLAN then opens the **PROJECT MANAGEMENT** dialog.

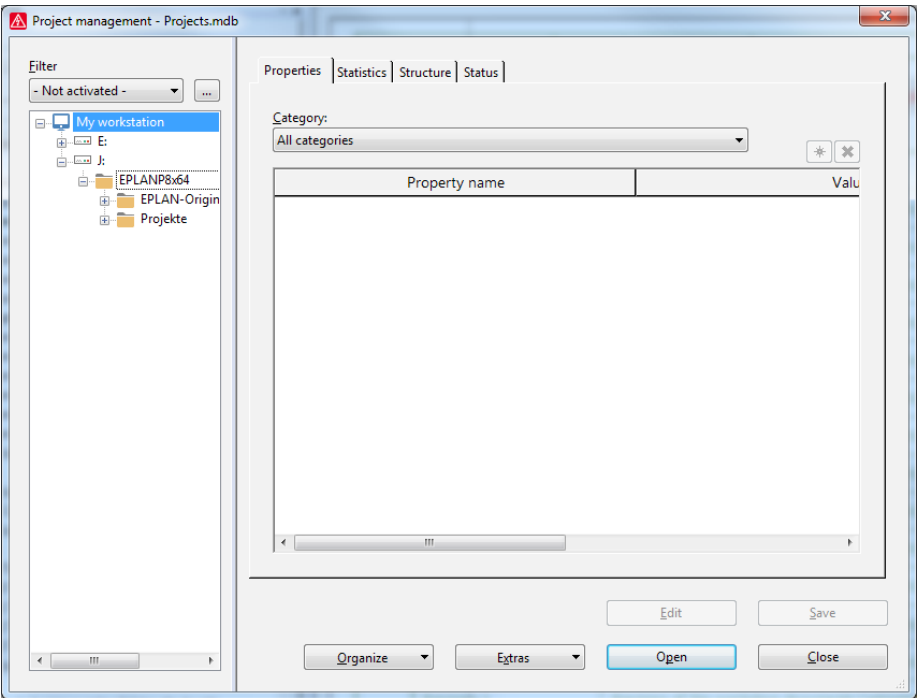


Fig. 7.129 Project management dialog

In the example, that would be the project database in the directory `\\EPLANP8x64\Projects\Gischel\Projects.mdb`.

The storage location of the project database can be modified in the **OPTIONS/SETTINGS/COMPANY/MANAGEMENT/PROJECT MANAGEMENT DATABASE** setting.

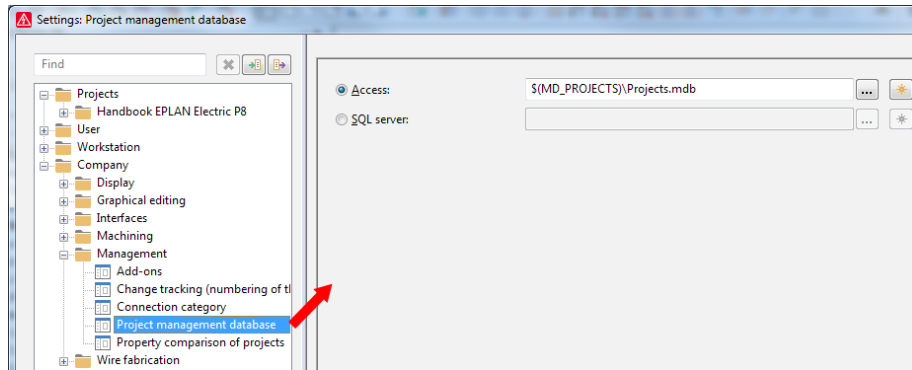


Fig. 7.130 Project management database storage path setting

The **PROJECT MANAGEMENT** dialog itself is divided into two key areas: the left area with an overview of the project directories and the projects on your own workstation (computer), or also on other workstations and the project tree.

Apart from the option to open or close the project tree by clicking the plus sign in front of the project directories, you can also use the **x** key of the numeric keypad to expand the project tree, or the **-** key to collapse it.

When closing project management, EPLAN saves the most recent representation of the directory tree and starts with the same representation when project management is called the next time.

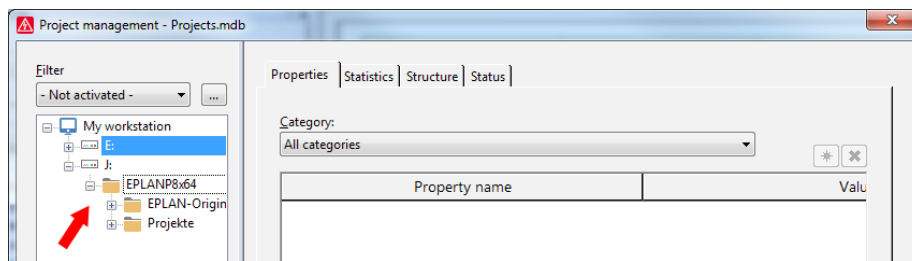


Fig. 7.131 Repeated opening of project management

The right area of the **PROJECT MANAGEMENT** dialog contains, at first sight, a display of information on the currently selected project or projects.

In the lower area, there are the buttons **ORGANIZE**, **EXTRAS**, **OPEN**, **CLOSE**, **EDIT**, and **SAVE**. I will now explain their most important functions.

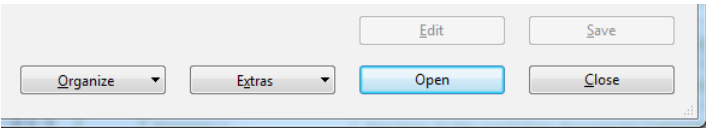


Fig. 7.132 Buttons in the Project management dialog

7.6.2 Project management buttons

7.6.2.1 Edit button

Use the **EDIT** button to configure the project settings of the currently selected project in the window on the left (project tree).

For example, this applies to the properties of the project (text entries) or structures of the various device groups. You should note that different settings, such as the page structure of the project, cannot be changed subsequently.

It is possible to select several projects and to edit their project settings collectively. This function is very similar to block editing and would apply the same changed entry to all projects selected.

- Example 1: Several projects selected, same settings for the selected projects in the user supplementary field 2.

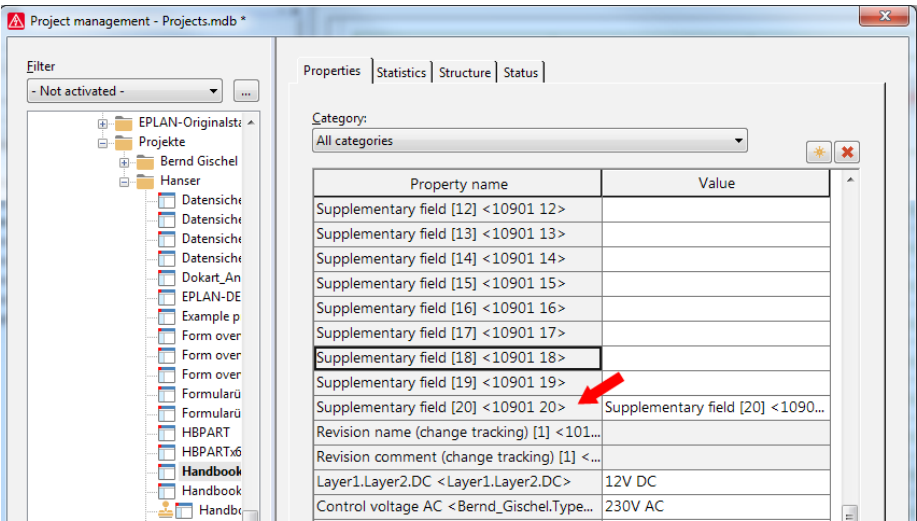


Fig. 7.133 Same properties of projects

- Example 2: Several projects selected, different settings for the selected projects in the user supplementary field 1.

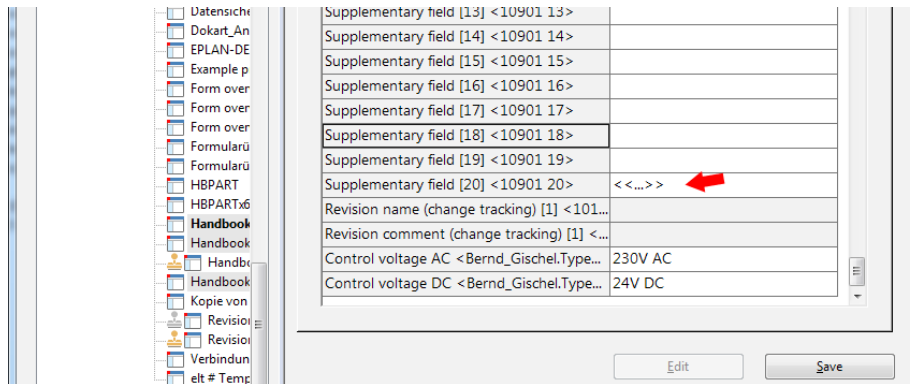


Fig. 7.134 Different properties of projects

While a change in the user supplementary field 2 is clear and unambiguous, a change in the user supplementary field 1 is daring, to say the least, because the various project entries of the respective projects cannot be differentiated clearly.

Different entries in property fields (not only in project properties, but also overall in EPLAN) can always be identified by the string of characters <<...>>!

At this point, there is no option to undo input after saving (UNDO function)! You should therefore use the *Block editing* function very carefully! Only when all selected projects are supposed to be assigned the same entry can you make an input here without worrying about it.

7.6.2.2 Save button

The SAVE button is explained quickly. After changing project settings and/or project properties, you save them using this button.

If you click a different project in the project tree (on the left) after making changes, the changes will be lost, however. EPLAN will prompt you with the **PROJECT PROPERTIES** dialog whether the changes are to be saved.

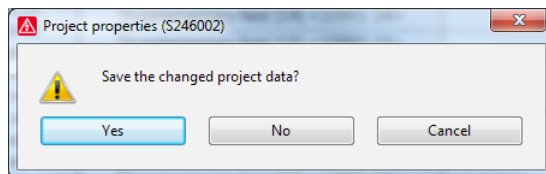


Fig. 7.135 Save project properties prompt

The prompt provides the usual options, such as **YES** (changes are saved), **NO** (changes are discarded, that is, you want to retain the original state of the project settings), or **CANCEL** (you continue to edit the current project settings).

7.6.2.3 Open button

Use the OPEN button to open and display the currently selected project(s). Project management is closed automatically in this case.

7.6.2.4 Close button

Use the CLOSE button to close project management. EPLAN jumps back to the graphical editor for further project editing of the current project.

Open projects (accessible in the page navigator) are represented in the project tree of project management in a visually different manner.

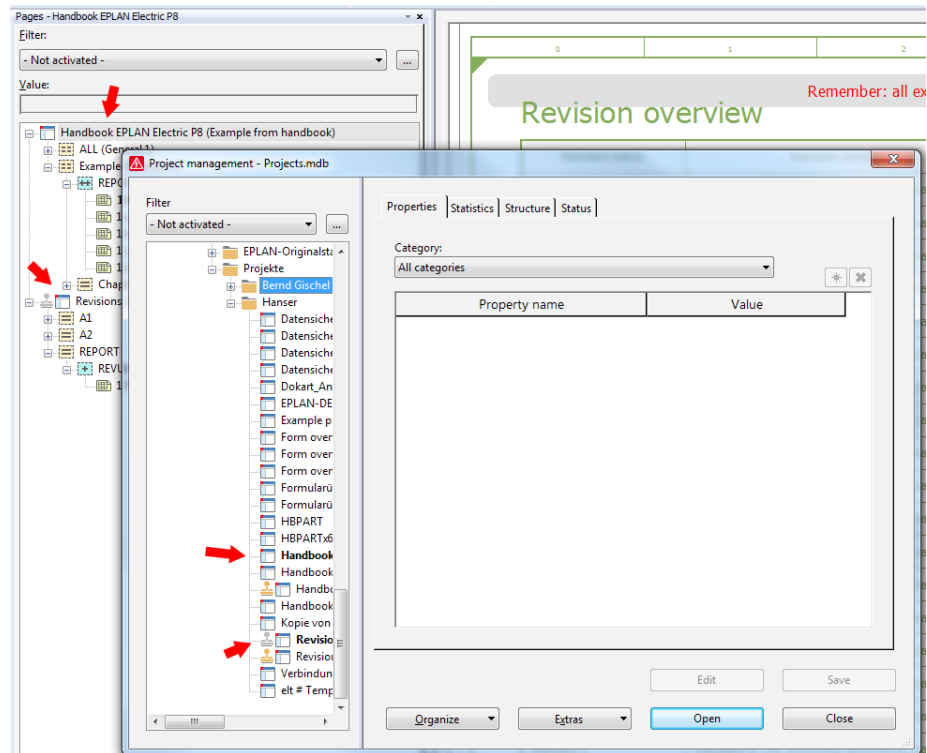


Fig. 7.136 Open projects in project management and in the page navigator

The default setting for the display is bold. In other words, projects already open in the page navigator appear in bold in the project tree of project management.

Projects already open in the page navigator cannot be opened again in EPLAN. If you still attempt it, EPLAN will display a warning that the project is already open.

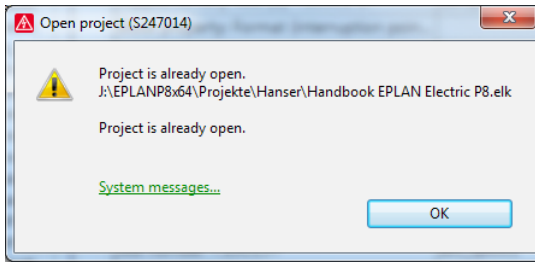


Fig. 7.137
Open project dialog

7.6.2.5 Organize button

In the **ORGANIZE** button, EPLAN provides further functions for the project editing and management of projects by way of an expandable popup menu.

In section (1) of the **ORGANIZE** popup menu, you find familiar functions to *open*, *move*, *rename*, *copy* or *delete* projects.

In section (2), new projects are generated from a basic project or template project (as already described).

Projects can be imported and/or exported in section (3) of the popup menu. These projects are assigned, or must have, the *.epj extension, and are available in the XML format.

Section (4) contains the functions *Compress* and/or *Reorganize* for projects.

The *Compress* function can be modified by means of configurable project parameters. The project parameters can be found in **SETTINGS/PROJECT/PROJECT NAME/MANAGEMENT/COMPRESS**. Here you can define many actions for compressing the respective project by using a scheme.

The *Reorganize* function is used to optimize the database. No other settings, such as via parameters, are possible here.

Section (5) contains the *Automated processing* function. This allows you to use ready-made schemes, or create schemes for automatic editing by calling the *New scheme* function.

This is done by way of the familiar procedure for schemes. The **SETTINGS...** button opens the **SETTINGS: AUTOMATED PROCESSING** dialog. Here you can, in the usual manner, generate and then define a new scheme with the desired actions. You can also import external schemes.

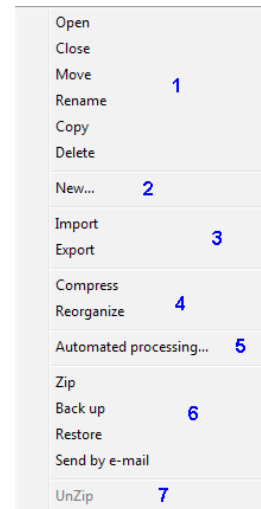


Fig. 7.138
Organize menu

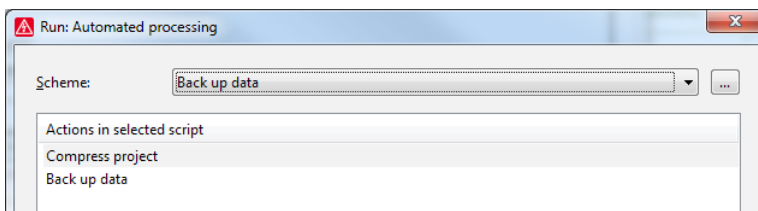


Fig. 7.139
Automated processing,
scheme selection